

# A Theory of the Visible Hand: Intermediation and Coordination in Markets for Relational Contracts

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## Abstract

The visible hand of intermediaries often replaces the invisible hand of markets in determining allocations. We develop repeated-game models showing when and why intermediaries enter relational contracts on both sides of a competitive market to coordinate assignments. We show that monitoring intermediaries can redress market misallocation when trust is lacking. However, they require an incentive rent, so in equilibrium buyers sort into decentralized bilateral exchange and centralized coordination. Their optimal choice depends on their demand volatility, need for specialized services, market tightness, market size, and reputation effects. Our theory explains the observed determinants and effects of professional service outsourcing.

Keywords: intermediation, relational contracts, markets

JEL: D23, L22, L24

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# 1 Introduction

For a large fraction of global economic activity, the *visible hand* of monitoring intermediaries supplants the *invisible hand* of markets in determining allocations. For example, professional service firms monitor workers and flexibly assign them to clients (Abraham and Taylor 1996). Global sourcing companies like Li & Fung coordinate and supervise buyer-supplier transactions (Belavina and Girotra 2012). Online platforms like Uber and Airbnb direct service provision by independent contractors and monitor its quality.<sup>1</sup> However, formal models showing why and when centralized monitors replace decentralized markets in resource allocation are illusive.<sup>2</sup>

In this paper, we analyze a simple repeated-game model of an exchange economy, in which market participants endogenously sort into decentralized exchange and centralized coordination by monitoring intermediaries. In our model, intermediaries cannot themselves produce, but can monitor a large number of producers. They enter multiple relational contracts on both sides of a market. We show that when trust is lacking, such intermediaries can redress misallocation that would otherwise arise in an anonymous matching market. They do so by aggregating fluctuating demand, coordinating assignments, keeping track of past producer performance, and excluding underperformers from future business. In short, they function like managers or bosses.

Our theory highlights a novel economic trade-off: In the presence of mistrust and demand fluctuations, monitoring intermediaries reduce both producer idleness and producer incentive rents. However, such intermediaries require an additional incentive rent. We show that buyers sort into intermediation and decentralized exchange in a unique steady-state equilibrium. We show that their optimal choice depends on their demand volatility, need for specialized services, market tightness, market size, and reputation effects. Unlike intermediation that mitigates search frictions or adverse selection, we show that this form of monitoring intermediation may *increase*

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<sup>1</sup>According to Wallis and North (1986), the private “transaction sector” was around 41% of U.S. GNP in 1970. Spulber (1996) estimates that intermediaries account for about 25% of U.S. GDP. Professional service firms alone account for 12% of US GDP (Berlingieri 2013). Ahn, Khandelwal and Wei (2011) show that intermediaries account for around 20% of China’s exports in 2005. Bernard, Grazi and Tomasi (2015) document that more than one-quarter of Italian exporters are intermediaries, and they account for over 10% of exports. See also books by Spulber (1999) and Krakovsky (2015).

<sup>2</sup>As explained in Section 1.1, existing models of intermediaries largely focus on search frictions (e.g., Rubinstein and Wolinsky 1987) and adverse selection (e.g., Biglaiser 1993). Models of monitoring intermediaries (e.g., Biglaiser and Friedman 1994; Belavina and Girotra 2012) are closer to ours in spirit but are very different in structure.

as communication frictions fall.

We then use the framework to understand the drivers and welfare consequences of intermediation in several empirical applications. We find that the model is consistent with empirical findings regarding the determinants and effects of professional service outsourcing (e.g. [Abraham and Taylor 1996](#); [Houseman 2001](#); [Garicano and Hubbard 2009](#); [Dube and Kaplan 2010](#); [Goldschmidt and Schmieder 2017](#)). We also show how the model can shed light on two other types of intermediary organizations: retail franchises and online platforms.

Section 2 develops our theory. We consider an exchange economy similar to [Shapiro and Stiglitz \(1984\)](#). In this economy, producers cannot be compelled to exert effort using written legal contracts, so buyers must motivate producers using future surplus in long-term relational contracts. They meet and enter contracts in frictionless matching markets without search delays or hidden payoff-relevant information. A new feature is added to this otherwise standard model: Buyers may have fluctuating demand. Since buyers with fluctuating demand have a shorter production horizon to offer, producers matched with such buyers require a higher incentive rent.

We show in this environment that intermediaries can reduce producer idleness and incentive rents by coordinating allocations. The key assumption needed for this result is that buyers observe only the efforts of a *single* matched producer, while intermediaries can observe the efforts of *multiple* matched producers. These intermediaries coordinate the assignment of producers to buyers and track their performance as producers serve different buyers. They can therefore offer relational contracts with longer production horizons to producers. In these contracts, producers never become idle or unmatched, so producer pay can be lowered without sacrificing producer motivation to exert effort.

The cost of intermediation, however, is that the buyer must pay an additional incentive rent to the intermediary. In other words, intermediation results in a form of double marginalization.<sup>3</sup>

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<sup>3</sup>In the classic literature in industrial organization, double marginalization occurs when both upstream and downstream parties possess market power and use restrictive linear contracts. This form of double marginalization can be eliminated by vertical integration of pricing decision rights, resale-price maintenance, or non-linear pricing ([Tirole 1988](#)). In our model, double marginalization instead arises from the non-verifiability of contract performance and the non-transferability of production and intermediation capabilities, like models of middleman margins ([Biglaiser and Friedman 1994](#); [Bardhan, Mookherjee and Tsumagari 2013](#)). Conventional remedies do not eliminate this form of double marginalization.

For this reason, buyers with sufficiently persistent demand prefer to directly contract. Bilateral contracts are also preferred if the producer market is slack and the opportunity cost of keeping a producer idle is therefore low.

Our two main propositions show that buyers sort into bilateral and intermediated contracts in a unique steady-state equilibrium. If the market is sufficiently tight, centralized intermediaries emerge as a contracting nexus that coordinates the assignment of producers to the subset of buyers whose business needs are sufficiently short-lived. The remaining buyers contract with producers in a sea of decentralized pairings.

We establish our main propositions by characterizing bilateral and intermediated contracts in Section 3 and deriving the steady-state equilibrium in Section 4. Three implications are then discussed. First, directly contracted producers have higher average incentive pay, more dispersed pay, higher separation rates, and higher idleness in equilibrium than intermediated producers. Second, the introduction of intermediaries into an economy benefits buyers with fluctuating demand, but reduces the pay of producers initially earning pay premiums. As such, intermediation is partly redistributive. Third, intermediation is profitable only if there is a sufficiently large market of buyers and producers. In other words, intermediation requires economies of scale. Potential market-level spillover effects from intermediation are also discussed.

Section 5 extends the model to analyze the relationship between intermediation, specialization, and reputation concerns. We show that intermediated producers are more specialized and that intermediation increases producer specialization. Moreover, buyers are more likely to choose intermediation if there are gains from producer specialization and if the intermediaries have reputation concerns arising from word-of-mouth communication about intermediary performance. In contrast to models of intermediaries based on search frictions and asymmetric information, we find that improvements in communication technologies may *increase* intermediation. The reason is that ease of communication may allow buyers to better coordinate on collective punishments against intermediaries who shirk. This lowers intermediary service fees, so intermediation becomes more attractive.

Section 6 shows that the model is consistent with a variety of recent findings in the empirical literature regarding professional service outsourcing. These findings include the firm-level and

market-level drivers of intermediation, as well as the effects of intermediation on both workers and the labor market as a whole. As such, our model provides a rich and realistic answer to a variant of the question first posed by Coase (1937) concerning the boundary between managerial coordination and market-based allocation. We also illustrate how our model can be applied to understand retail franchises and online platforms.

## 1.1 Related Theories and Contributions

The idea that trust and relationships are important in economic exchange has a long pedigree: in law, Macaulay (1963) and Macneil (1978) emphasize the relational nature of exchange contracts; in anthropology, Geertz (1962, 1978) highlight relational aspects of the bazaar economy; in sociology, Granovetter (1985) underscores the embeddedness of economic exchange; in strategy, Dyer and Singh (1998) studies long-term relationships between firms.<sup>4</sup> The novelty here is to explicitly show how a need for trust leads to the emergence of allocative and monitoring intermediaries. Our theory connects to existing work on relational contracts, intermediation, market institutions, and the theory of the firm.

First, we build on a game-theoretic literature on relational contracts in matching markets (Shapiro and Stiglitz 1984; MacLeod and Malcomson 1989, 1998; Yang 2008; Board and Meyer-ter Vehn 2015; Fahn 2017; Powell 2019; Fahn and Murooka 2022; Li 2022). The novelty is to introduce *intermediaries* into frictionless markets for relational contracts. We show that by aggregating fluctuating demand and coordinating assignments, intermediaries can ease the tension between incentive provision and flexible allocations, and thereby enable more trade and specialization. In related work, Board (2011) shows that this tension can lead buyers to restrict their number of trading partners. Andrews and Barron (2016) show that this tension can cause dynamic allocations to depend on payoff-irrelevant past performance. Li and Powell (2020) highlight that a similar tension can lead agents to interact in multiple activities.<sup>5</sup>

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<sup>4</sup>For further examples, see also Powell (1990); Bernstein (1992, 2016); Fafchamps (2004); Greif (2006); Gibbons and Henderson (2012).

<sup>5</sup>For surveys of recent work on relational contracts, see MacLeod (2007), Malcomson (2013), Gil and Zananone (2017), and Macchiavello and Morjaria (2023). There are also two related models of relational contracting intermediaries. Fong and Li (2017) show that relational contracts can be deepened by the presence of a non-strategic supervisor carrying out subjective performance reviews. Troya-Martinez and Wren-Lewis (2023) explore a

Second, our work is related to the literature on intermediation. One strand of this literature studies middlemen who overcome search frictions.<sup>6</sup> Another studies intermediaries who overcome adverse selection.<sup>7</sup> Very few papers purely examine how intermediaries redress moral hazard. Most relatedly, [Biglaiser and Friedman \(1994\)](#) and [Bardhan, Mookherjee and Tsumagari \(2013\)](#) study models in which middlemen with reputations have a larger incentive to monitor and are in a better position to learn about quality than an ordinary buyer does because they buy a larger proportion of the producers' goods. In contrast, middlemen do not have reputations in our baseline model. They improve performance simply by entering multiple relational contracts on both sides of the market. In the supply chain management literature, [Belavina and Girotra \(2012\)](#) develop a related model of relational intermediaries with two buyers, two sellers, and non-transferable utilities. However, they do not provide results regarding the determinants and welfare consequences of intermediation.

Third, our work connects to a game-theoretic literature on market institutions. Famously, [Milgrom, North and Weingast \(1990\)](#) show that private judges in medieval trade can create trust with less observability than reputation systems. [Greif, Milgrom and Weingast \(1994\)](#) highlight the role of merchant guilds who mechanically announce boycotts of a city who cheats its traders.<sup>8</sup> Our paper follows a similar logic. However, our focus is not the historical role of legal and regulatory institutions in trade. We instead view the problem of trust as pervasive despite the development of modern legal institutions. We argue that it leads to monitor coordination in many sectors of the economy.<sup>9</sup>

Fourth and finally, our work can be viewed as formalizing ideas from an early strand of the theory of the firm. In a classic article, [Alchian and Demsetz \(1972\)](#) proposed that the

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model where a manager may receive kick-backs and show that such managers can improve relational contracts in environments with commitment difficulty. In the supply chain management literature, [Tunca and Zenios \(2006\)](#) model the choice between bilateral relational contracts and the use of a procurement auction, but do not consider the role of intermediaries.

<sup>6</sup>See, for example, [Rubinstein and Wolinsky \(1987\)](#), [Gehrig \(1993\)](#), [Yavaş \(1994, 1996\)](#), [Rust and Hall \(2003\)](#), [Antras and Costinot \(2011\)](#), [Wright and Wong \(2014\)](#), [Nosal, Wong and Wright \(2015\)](#), [Nosal, Wong and Wright \(2019\)](#), and [Rhodes, Watanabe and Zhou \(2021\)](#).

<sup>7</sup>See, for example, [Biglaiser \(1993\)](#), [Lizzeri \(1999\)](#), [Glode and Opp \(2016\)](#), and [Biglaiser and Li \(2018\)](#).

<sup>8</sup>See also [Greif \(1993, 2006\)](#), [Calvert \(1995\)](#), [Ramey and Watson \(2002\)](#), [Fafchamps \(2004\)](#), and [Budish \(2023\)](#).

<sup>9</sup>[Henderson and Churi \(2019\)](#) offer a related perspective and traces the history of innovation and trust from medieval guides, early corporations, New Deal administrative agencies, and Internet-age companies such as Uber.

essence of the firm is the presence of a central contracting party acting as a monitor who can exclude team members from future participation in the team. [Jensen and Meckling \(1976, pp. 310–311\)](#) expanded on this view and suggested that “contractual relations are the essence of the firm, not only with employees but with suppliers, customers, creditors, and so on.” In a further elaboration, [Demsetz \(1988\)](#) proposed that continuity of association, specialization, and managerial direction were key attributes of firm-like organization.<sup>10</sup> Our framework clarifies these ideas by deriving conditions under which specialized monitor-coordinators emerge as nexuses of *relational* contracts.

The perspective propounded here is complementary to other strands of the theory of the firm. For instance, the *transaction cost* approach proposed by [Williamson \(1971, 1975, 1985\)](#) catalogs forces such as asset specificity, bounded rationality, bargaining frictions, and contract-writing costs that drive the “make-or-buy” decision.<sup>11</sup> The *property rights* approach inaugurated by [Grossman and Hart \(1986\)](#) offers a formal framework to analyze the incentive effects of asset ownership.<sup>12</sup> Their approaches are orthogonal to ours since they focus on characterizing the *integration* choices of firms. We instead focus on analyzing *intermediation* choices in exchange.

## 2 Model

**Basics.** Time is discrete and infinite,  $t \in \{0, 1, \dots\}$ . There are a unit mass of buyers indexed by  $i$  who demand services. There is a continuum of producers indexed by  $j$  who provide services. The measure of producers is determined by endogenous entry at the start of each period subject to entry cost  $C$ , which represents the producers’ training or opportunity cost. There is a finite number  $K$  of intermediaries indexed by  $k$  who neither directly demand nor provide services. All players are infinitely-lived and have a common discount factor  $\delta \in (0, 1)$ .

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<sup>10</sup>See also [Cheung \(1983\)](#), [Spulber \(1996\)](#), [Rajan and Zingales \(1998, 2001\)](#), and [He \(2016\)](#).

<sup>11</sup>For formal models that build on this approach, see [Bajari and Tadelis \(2001\)](#); [Tadelis \(2002\)](#); [Hart and Moore \(2007, 2008\)](#); [Levin and Tadelis \(2010\)](#); [Tadelis and Williamson \(2012\)](#); [Wernerfelt \(2015, 2016\)](#).

<sup>12</sup>See, e.g., [Gibbons \(2005\)](#) for an comprehensive treatment of related formal theories.

**Demand realization.** After the endogenous entry of producers, the service demand of each buyer  $i$ , denoted  $d_{it} \in \{0, 1\}$ , is realized and publicly observed.<sup>13</sup> Demand  $d_{it}$  is redrawn at the beginning of each period  $t$  following a Markov process. The demand-switching probabilities are each buyer's publicly known type  $\alpha_i = (\alpha_{1i}, \alpha_{0i})$ . With probability  $\alpha_{1i}$ , buyer  $i$ 's demand switches from 1 to 0. With probability  $\alpha_{0i}$ , buyer  $i$ 's demand switches back from 0 to 1. The distribution of buyers types is a distribution  $F$  on  $[0, 1] \times [0, 1]$ .

**Matching.** Both buyers and intermediaries can offer contracts to producers in a producer market. In addition, buyers can offer contracts to intermediaries in an intermediary market. Each contract is a contingency plan specifying compensation, effort levels, and the probability of continuation when demand switches. Following [Board and Meyer-ter Vehn \(2015\)](#), matching is frictionless, meaning that all offers are accepted subject to participation constraints.

In the producer market, matching is random and anonymous in that all unmatched producers have the same probability of being matched with a given buyer or intermediary regardless of their histories; matched producers do not receive contract offers.<sup>14</sup> In the intermediary market, matching is not anonymous, so matching probabilities may depend on the observable history of an intermediary. Since intermediaries cannot produce by themselves, they meet the requirements of contracts with buyers by entering contracts with producers.

We assume that an intermediary can match with a continuum of buyers and producers. However, in any period, buyers can match with at most one producer or intermediary, while producers can match with at most one buyer or intermediary.<sup>15</sup> Matching in intermediary market precedes matching in the producer market, so intermediaries can always fulfill the service demands of their matched buyers if there is an excess of producers.

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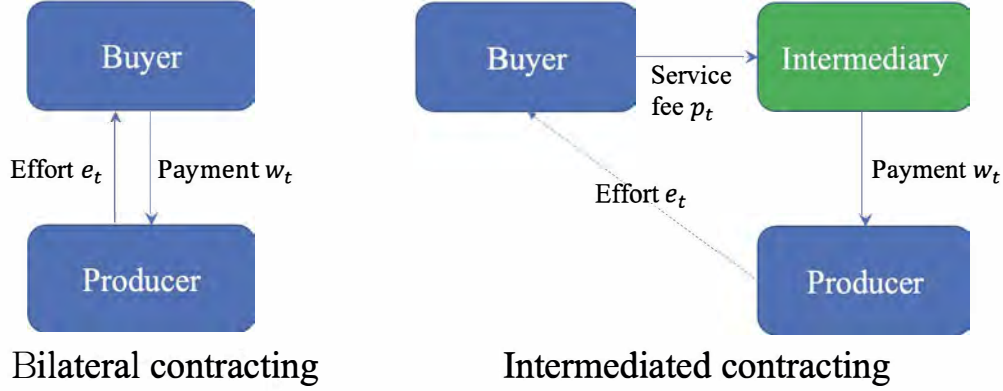
<sup>13</sup>To focus on the relational incentives, we abstract from any adverse selection problem in the model. Therefore, when contracting with buyers, producers know whom they are dealing with and have the correct expectation of how the relationships will go.

<sup>14</sup>The assumption of random and anonymous matching simplifies analysis but is not important. Similar results can be obtained under the assumption that buyers and intermediaries first offer contracts to unmatched producers with whom they had previously matched. Relatedly, [Board and Meyer-ter Vehn \(2015\)](#) analyzes relational contracts in a frictionless matching market where matched producers may receive on-the-job offers. They show this leads to heterogeneous productivity across otherwise identical firms. We rule out this possibility for simplicity.

<sup>15</sup>This assumption rules out the possibility that producers and buyers become intermediaries themselves.



Figure 1: Illustration of bilateral and intermediated contracts



**Bilateral contracts.** If a buyer matches with a producer, the remainder of the period proceeds as follows. First, the buyer makes a payment  $w_t \geq 0$  to the producer.<sup>16</sup> The producer then chooses an effort level, denoted by  $e_t \in \{0, 1\}$ . The cost of effort is given by  $c(e_t)$ , where  $c(0) = 0$  and  $c(1) = c > 0$ . The effort generates an output for the buyer only if demand is positive, so  $y_{it} = y d_{it} e_t$ . Effort and output are observable by the buyer but are not verifiable by a court. We assume that  $y > \frac{c}{\delta} + (1 - \delta)C$ , so that for buyers with unchanging and positive demand, there is always enough surplus to incentivize producer effort.

**Intermediated contracts.** If a buyer matches with an intermediary, she pays a service fee  $p_t \geq 0$  to the intermediary. The intermediary chooses to assign one or none of its producers to the buyer. The intermediary pays  $w_t$  to the producer. The producer then exerts costly effort  $e_t$  and produces output  $y_{it}$  for the buyer.<sup>17</sup> Effort and output are observable by both the buyer and the intermediary, but are not verifiable by a court.

<sup>16</sup>Here we do not allow for ex-post bonus payments, which play an important role in [MacLeod and Malcomson \(1998\)](#) and [Levin \(2003\)](#). This assumption is not important: There is an excess of producers, so buyers in our model have all the bargaining power. It can then be shown that for any contract with bonuses, there is a weakly more profitable contract for buyers without bonuses. Intuitively, the buyer prefers to pay the producer for effort  $e_{t-1}$  at the latest possible moment before  $e_t$ . It is therefore weakly profitable to shift bonus  $b_{t1}$  into next period's pay  $w_t$ . See [Board and Meyer-ter Vehn \(2015\)](#).

<sup>17</sup>Similarly, we do not allow for ex-post service fees. This is unimportant for the same reason that ex-post bonuses are assumed away; buyers can immediately rematch with new intermediaries, and contracts with ex-post service fees are weakly less profitable for buyers.

**Separation.** Either party in a match can choose to terminate their contract and separate from each other both after demand realization and after production. If separation occurs after demand realization, both parties can participate in the producer or intermediary market matching in the current period. However, if separation occurs after production, they need to wait till the next period to rematch.

**Payoffs.** In a bilateral contract, the buyer's payoff is  $y_{it} - w_t$  in each period  $t$ . In an intermediated contract, the buyer's payoff is  $y_{it} - p_t$ . The intermediary's payoff per service demand is  $p_t - w_t$ . The producer's payoff is  $w_t - c(e_t)$ .

**Remark.** This model has two key features that are not present in standard models of relational contracting in frictionless matching markets, such as [Shapiro and Stiglitz \(1984\)](#). First, we allow buyers to have fluctuating demand. Second, we introduce intermediaries who can enter relational contracts with a multitude of agents on both sides of the market. These intermediaries do not have intrinsic demand or productive ability. What they can do, however, is to match with a large set of buyers and producers and monitor the performance of all of their matched producers. As we shall show, when demand is volatile and the cost of idleness is high, intermediaries can sustain cheaper relational contracts on both sides of the market by reassigning their producers across buyers. Buyers may therefore prefer intermediation rather than bilateral contracting.

### 3 Bilateral and Intermediated Relational Contracts

In this section we characterize optimal bilateral and intermediated relational contracts. We then analyze the buyer's choice between bilateral and intermediated contracts. This analysis explains why and when intermediation may be preferred over bilateral contracting.

#### 3.1 Bilateral Relational Contracts

To analyze bilateral relational contracts, we take the perspective of a single buyer  $i$ . Since producers are homogeneous, we omit the  $j$  subscript. We assume that the producer's pre-matching

continuation value is exogenously given as  $\bar{U} > 0$ . We endogenize  $\bar{U}$  in Section 4.

We say that strategies under a relational contract are *contract-specific* if they do not depend on the player's identity, calendar time, or any history outside the current contract. A contract is *stationary* if strategies are time-invariant functions of the buyer's demand realizations. A contract is offerer-optimal if it yields the highest possible surplus for the party offering the contract. We restrict our attention to buyer-optimal, contract-specific, stationary contracts in which producer's effort level is one if and only if  $d_{it} = 1$ .<sup>18</sup> These contracts must also satisfy the following two conditions. First, on the equilibrium path, parties within a match always choose to continue their relationship immediately after production. Second, off the equilibrium path, deviations are punished in the harshest possible way. These assumptions are standard in the literature (MacLeod and Malcomson, 1998; Baker, Gibbons and Murphy, 2002; Board and Meyer-ter Vehn, 2015).

Let  $C_i^B = (w_{1i}, w_{0i}, \beta_i)$  denote a contract-specific, stationary relational contract offered by buyer  $i$  directly to a producer. In this contract,  $w_{1i}$  is the payment when  $d_{it} = 1$ ,  $w_{0i}$  is the payment when  $d_{it} = 0$ , and  $\beta_i \in [0, 1]$  is the probability that buyer  $i$  stays with the producer when the buyer's demand switches from one to zero. The time subscript is dropped since we focus on stationary contracts.

Under a bilateral contract, the buyer motivates the producer to exert effort using credible promises of future surplus from their contractual relationship. If the buyer deviates from the specified payment, the producer exerts no effort and separates from the buyer after production with probability one. If the producer deviates from the specified effort, the buyer separates from the producer after production with probability one. Since the buyer has fluctuating demand, the retention probability  $\beta_i$  determines the expected duration of the relationship and therefore affects the level of payments needed to incentive the producer.

If  $\beta_i > 0$ , the post-matching continuation payoffs for the producer when  $d_{it} = 1$  and  $d_{it} = 0$

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<sup>18</sup>Focusing on stationary relational contracts is without loss when imposing pairwise stability (or “bilateral efficiency” in Board and Meyer-ter Vehn (2015)) on the equilibrium concept. A pairwise stable relational contract is a Pareto-optimal contract for parties in a match when they take their outside options as given. See Li (2022) for a more detailed discussion on how non-stationary relational contracts may be optimal in equilibrium when the pairwise stability restriction is relaxed.

are, respectively, given by

$$U_{1i} = w_{1i} - c + \delta \left[ (1 - \alpha_{1i})U_{1i} + \alpha_{1i}(\beta_i U_{0i} + (1 - \beta_i)\bar{U}) \right], \quad (1)$$

and

$$U_{0i} = w_{0i} + \delta \left[ \alpha_{0i}U_{1i} + (1 - \alpha_{0i})(\beta_i U_{0i} + (1 - \beta_i)\bar{U}) \right]. \quad (2)$$

The relevant incentive constraints for the producer are as follows:

$$U_{1i} \geq w_{1i} + \delta \bar{U}, \quad (\text{P-IC-e})$$

$$U_{1i} \geq \bar{U}, \quad (\text{P-IC1})$$

$$U_{0i} \geq \bar{U}. \quad (\text{P-IC0})$$

Constraint (P-IC-e) requires the producer to choose effort over shirking when the service is needed. Constraints (P-IC1) and (P-IC0) require that the producer remain with the current buyer when the demand is 1 or 0, respectively. If  $\beta_i = 0$ , equation (2) and constraint (P-IC0) do not apply, as the buyer immediately separates from the producer when demand is zero.

For the buyer, the post-matching continuation payoffs when  $d_{it} = 1$  and  $d_{it} = 0$  are

$$\Pi_{1i} = y - w_{1i} + \delta \left[ (1 - \alpha_{1i})\Pi_{1i} + \alpha_{1i}(\beta_i \Pi_{0i} + (1 - \beta_i)\bar{\Pi}_{0i}) \right],$$

and

$$\Pi_{0i} = -w_{0i} + \delta \left[ \alpha_{0i}\Pi_{1i} + (1 - \alpha_{0i})(\beta_i \Pi_{0i} + (1 - \beta_i)\bar{\Pi}_{0i}) \right].$$

where  $\bar{\Pi}_{1i}$  and  $\bar{\Pi}_{0i}$  is the value of the buyer's pre-matching continuation values when  $d_{it} = 1$  or  $d_{it} = 0$ , respectively. Since there is an excess of producers in the frictionless matching market, the buyer can always successfully find a match, so  $\bar{\Pi}_{1i} = \Pi_{1i}$ .

The relevant incentive constraint for the buyer are:

$$\Pi_{1i} \geq \delta(\alpha_{1i}\bar{\Pi}_{0i} + (1 - \alpha_{1i})\bar{\Pi}_{1i}), \quad (\text{B-IC-w})$$

$$\Pi_{1i} \geq \bar{\Pi}_{1i}, \quad (\text{B-IC1})$$

$$\Pi_{0i} \geq \bar{\Pi}_{0i}. \quad (\text{B-IC0})$$

Constraint (B-IC-w) ensures that the buyer honors the payment to the producer. Constraints (B-IC1) and (B-IC0) reflect the buyer's desire to retain the producer when there is a demand or not, respectively. As before, if  $\beta_i = 0$ , the term  $\Pi_{0i}$  and constraint (B-IC0) do not apply, as the buyer would immediately separate from the producer.

The optimal bilateral contract is obtained by choosing  $w_{1i}$ ,  $w_{0i}$ , and  $\beta_i$  to maximize  $\Pi_{1i}$ , subject to (B-IC-w), (B-IC1), (P-IC-e), (P-IC1), as well as (B-IC0) and (P-IC0) if  $\beta_i > 0$ .

**Lemma 1.** *Suppose an optimal bilateral relational contract exists. Under this contract, if*

$$\frac{(1-\delta)\bar{U}}{c} > \frac{\alpha_{0i}}{1-\alpha_{1i}}, \quad (3)$$

*then the buyer pays*

$$w_{BS}(\alpha_i) = \left( \frac{1}{\delta} \frac{1}{1-\alpha_{1i}} \right) c + (1-\delta)\bar{U}. \quad (4)$$

*when demand is one and separates from the producer when demand becomes zero. Otherwise, the buyer pays*

$$w_{BI}(\alpha_i) = \left( \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i}) + \frac{\delta}{1-\delta}\alpha_{0i}} \right) c + \left( \frac{1 + \frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i}) + \frac{\delta}{1-\delta}\alpha_{0i}} - \delta \right) \bar{U} \quad (5)$$

*when demand is one, remains matched with the idle producer when demands switches to zero, and pays zero until demand switches back to one.*

*Proof.* All omitted proofs are in the appendix. □

Lemma 1 shows that the optimal bilateral contract features a buyer who either always separates from a producer or always retains him when their demand switches to zero. According to Equation (3), bilateral contracts with separation dominates bilateral contracts with idleness when (a) the producer's continuation value when unmatched  $\bar{U}$  is high, (b) the buyer has a smaller  $\alpha_{0i}$ , so

a longer period of idleness is expected, and (c) the buyer has a large  $\alpha_{1i}$ , so a shorter period without idleness is expected.

The payment to producers in a bilateral contract,  $w_B(\alpha_i) \equiv \min\{w_{BS}(\alpha_i), w_{BI}(\alpha_i)\}$ , is increasing in  $\alpha_{1i}$ . This is because when business needs are shorter-lived, the producer faces a higher chance of either separating or becoming idle, so the future surplus in the relationship is smaller. A higher incentive rent is therefore needed to incentivize producer effort.

### 3.2 Intermediated Relational Contracts

Under intermediated contracts, buyers delegate to intermediaries the responsibility of motivating and monitoring producers. The intermediary fulfills the buyers's demand by entering relational contracts with a large number of producers and assigning producers to buyers according to demand realization.

We first analyze the contract that intermediaries offer to producers. To meet buyer demand, we assume that each intermediary offers intermediary-optimal, contract-specific, and stationary contracts to producers. As shown in Section 3.1, the terms in an optimal bilateral contract with producer hinge on the buyer's demand-switching probabilities. Unlike buyers, however, intermediaries face constant demand for services and therefore have constant demand for effort from its matched producers. The reason is that each intermediary randomly matches with a continuum of buyers drawn from the same distribution, so by the law of large numbers, the intermediaries face total demand from buyers that is constant over time. Anticipating this stable demand for services, the measure of producers that each intermediary contracts with is equal to the expected measure of demand realizations, and each matched producer is asked to exert effort in every period. Therefore, by the logic of Lemma 1, the compensating payment for the producer is  $w_{1i} = w_M$  in every period, where

$$w_M = \frac{c}{\delta} + (1 - \delta)\bar{U}. \quad (6)$$

We next consider how producers are assigned to buyers in each period under the intermediated contract. Note that buyers are indifferent between any assignment of producers where the assigned

producer exerts effort, since producers are identical in our model. Producers are also indifferent between any assignment of buyers where the intermediaries offer the same level of payments. Since intermediaries require producers to exert effort and provide the same compensating payment in every period, buyers and producers have the same payoffs in any assignment where producers are matched with buyers with positive demand in every period. There are an infinite number of such assignments, and any such assignment is optimal.

We now characterize the contract that buyers offer to intermediaries. Let  $C_i^M = (p_{1i}, p_{0i}, \beta_i)$  denote a buyer-optimal, contract-specific, and stationary intermediated relational contract offered by a buyer to an intermediary. Here  $p_{1i}$  and  $p_{0i}$  are the service fees when the buyer needs and does not need the service, respectively. If the buyer deviates from the specified service fee, the intermediary does not assign a producer to the buyer and separates from the buyer. If instead the producer assigned by the intermediary deviates from the specified effort, the buyer separates from the intermediary after production. Since matching is not anonymous in the intermediary market, the buyer never chooses to match with the defaulted intermediary again in the future.<sup>19</sup>

Under  $C_i^M$ , the post-matching continuation payoffs for the intermediary in a buyer-intermediary match, when the service is and is not needed, respectively, are

$$V_{1i} = p_{1i} - w_M + \delta \left[ (1 - \alpha_{1i})V_{1i} + \alpha_{1i}(\beta_i V_{0i} + (1 - \beta_i)\bar{V}) \right], \quad (7)$$

and

$$V_{0i} = p_{0i} + \delta \left[ \alpha_{0i}V_{1i} + (1 - \alpha_{0i})(\beta_i V_{0i} + (1 - \beta_i)\bar{V}) \right], \quad (8)$$

where  $\bar{V}$  is the value of an intermediary's continuation value after separating from a buyer. The relevant incentive constraints for the intermediary, similar with those for a producer, are

$$V_{1i} \geq p_{1i} + \delta \bar{V}, \quad (\text{M-IC-w})$$

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<sup>19</sup>We assume that in the intermediary market, buyers randomly match with an intermediary who has not breached the relational contracts with them. Specifically, let  $\mathcal{K}$  be the set of all intermediaries and  $\mathcal{K}_{i,t}$  be the subset who have interacted with buyer  $i$  and breached the relational contract with  $i$  before period  $t$ . Should a buyer become unmatched and match with an intermediary in the intermediary market in period  $t$ , she randomly matches with one of the intermediaries from the set  $\mathcal{K} \setminus \mathcal{K}_{i,t}$ .

$$V_{1i} \geq \bar{V}, \quad (\text{M-IC1})$$

$$V_{0i} \geq \bar{V}. \quad (\text{M-IC0})$$

Note that  $\bar{V} = 0$  on the equilibrium path. If an intermediary separates from a buyer, it cannot match with a new buyer, because all other potential buyers are matched with some intermediary and will not become unmatched on the equilibrium path.

For the buyer, the continuation payoffs and incentive constraints are the same as in the bilateral contract, except that the payments  $w_{0i}$  and  $w_{1i}$  to the producer are replaced with service fee payments  $p_{0i}$  and  $p_{1i}$  to the intermediary. For concision, we omit these conditions, which simply repeat (B-IC-w), (B-IC-1), and (B-IC0). The optimal intermediated contract maximizes  $\Pi_{1i}$  subject to these incentive compatibility constraints.

**Lemma 2.** *Suppose an optimal intermediated contract exists. Under this contract, the buyer always retains the intermediary, pays zero service fees to the intermediary when there is no demand, and when there is demand, she pays the intermediary a service fee equal to*

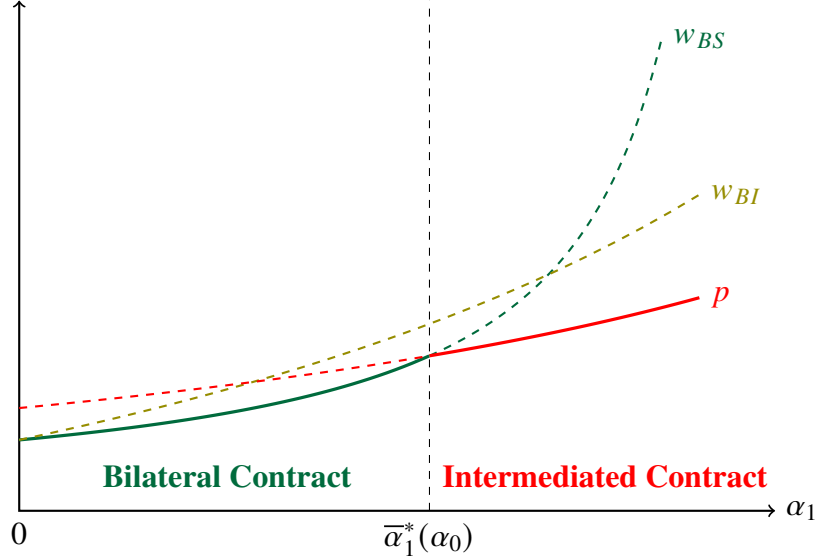
$$p(\alpha_i) = \left( \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_{0i}}{(1 - \alpha_{1i}) + \frac{\delta}{1-\delta} \alpha_{0i}} \right) w_M. \quad (9)$$

Lemma 2 shows that the cost of intermediated contract is a form of double marginalization. Note that  $p(\alpha_i)$  can be rewritten as the product of  $\lambda(\alpha_i) = \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_{0i}}{(1 - \alpha_{1i}) + \frac{\delta}{1-\delta} \alpha_{0i}}$  and the payment to the producer  $w_M$ . Here  $\lambda(\alpha_i)$  can be thought of as an intermediary markup. Furthermore, as shown in Equation (6),  $w_M$  is elevated above the cost of effort to the producer. In other words, both the intermediary and producer are paid rents so that both are incentivized to honor their contractual obligations.

The benefit of intermediated contract is that the payment to the producer is lower than the payment under bilateral contracts, since the intermediary can smooth demand across its buyers. To see this, note that  $w_B(\alpha_i) \geq w_M$  for all  $\alpha_i$ , where equality holds if and only if  $\alpha_{1i} = 0$ .



Figure 2: Service cost under bilateral and intermediated contracts



Note:  $\alpha_1$  is the probability that the buyer's demand switches from 1 to 0.

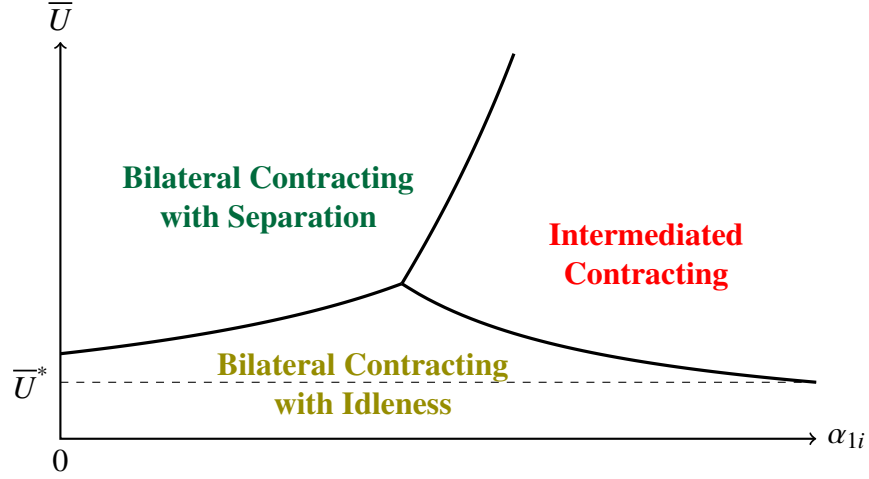
### 3.3 Optimal Contractual Choice

Having characterized bilateral and intermediated contracts, we now characterize when buyers choose intermediation. We focus on buyers who have the same  $\alpha_{0i}$ , which is the rate at which a buyer's demand changes from zero to one. We ask how the optimal contractual choice depends on  $\alpha_{1i}$ , the rate at which the buyer's demand changes from one to zero, and  $\bar{U}$ , the producer's entry cost, which determines whether the producer market is tight and whether it is easy for producers to rematch.

To compare the two arrangements, it suffices to compare the payment  $w_B(\alpha_i)$  under bilateral contracting and the service fee  $p(\alpha_i)$  under intermediation. This is because the buyer pays nothing when their demand is zero and can always match with a producer when her demand is one.

Figure 2 provides a graphical illustration of this comparison. For a buyer with stable demand, i.e., for whom  $\alpha_{1i} = 0$ , intermediated contracting is strictly more expensive than bilateral contracting because of double marginalization. To obtain high-quality services, the buyer needs to pay additional rent to the intermediary. However, there is no benefit to intermediation, since

Figure 3: Optimal contractual choice given model parameters



demand is stable, so the producer is paid the same incentive rent under bilateral contracting.

The advantage of intermediation over bilateral contracting becomes larger when business needs are short-term. As  $\alpha_{1i}$  becomes larger, producers must be paid elevated payments in order for them to exert effort. Since intermediaries can reassign producers across buyers based on demands so producers neither separate nor become idle, the cost of incentivizing producers is lowered. To see this mathematically, note that  $w_{BS}(\alpha_i)$  approaches infinity as  $\alpha_{1i}$  approaches one. Furthermore,  $p(\alpha_i)$  increases in  $\alpha_{1i}$  less steeply than  $w_{BI}(\alpha_i)$  if  $c$  is relatively small and  $\bar{U}$  is relatively large. Therefore, when rematching is easy, the intermediary operates a more cost-efficient internal producer market by having long-standing relational contracts on both sides of the market.

Figure 3 graphically shows the optimal contracts as a function of  $\alpha_{1i}$  and  $\bar{U}$ . In this figure, provided  $\bar{U} > \bar{U}^*$ , then there exists a cutoff value such that intermediated contracts dominates if and only if  $\alpha_{1i}$  is sufficiently large. If  $\bar{U} < \bar{U}^*$ , then the producer's pre-matching continuation value is low, so it is optimal to retain them and keep them idle until demand returns.

The following proposition formalizes this result.

**Proposition 1.** *Take any set of buyers  $i$  with the same  $\alpha_{0i}$  for whom an optimal contract exists. There exists  $\bar{U}^* > 0$  such that:*

1. *If  $\bar{U} < \bar{U}^*$ , a bilateral contract is optimal for all  $i$  in this set;*

2. If  $\bar{U} > \bar{U}^*$ , there exists  $\alpha_1^*(\alpha_{0i}) \in (0, 1)$  such that an intermediated contract is optimal if and only if  $\alpha_{1i} \geq \alpha_1^*(\alpha_{0i})$ .

Proposition 1 can be viewed the answer to a variant of the question first posed by Coase (1937): Why and under what conditions should we expect centralized allocators to emerge in decentralized markets? The above result shows that intermediation dominates bilateral pairings when business needs are short-term and the continuation value of an unmatched producer is high. Moreover, intermediation can dominate bilateral pairings even though buyers and producers in the model can frictionlessly meet, possess no hidden information about capabilities, and can freely enter contracts. Our result therefore implies that centralized allocators may emerge in the absence of many types of transaction costs, including search, bargaining, and contract-writing costs that are previously emphasized in the theory of the firm.

The model's prediction that buyers with frequent demand are more likely to disintermediate is consistent with findings in related literature. Williamson (1985) attributes this tendency to administrative and bargaining costs in repeated transactions. Wernerfelt (2015, 2016) attributes it to switching costs. In our model, the cost of intermediation is instead a price premium charged by the intermediary to ensure its performance as an aggregator, monitor, and contract enforcer.

## 4 Steady-State Equilibrium

In this section, we show that there exists a unique steady-state equilibrium. We then explore the model's empirical implications. First, we compare the outcomes of intermediated and bilaterally contracted producers in equilibrium. Second, we compare the outcomes and welfare of buyers and producers in economies with and without intermediaries. Third, we study how intermediation choices depends on market size.

### 4.1 Deriving the Equilibrium

We say that the economy is in a steady state when (1) the number of producers in the economy and the distribution of contracts in the matching market are unchanging across periods and (2)

each buyer's demand evolves according to its long-run distribution. By the properties of Markov processes, the steady-state probability that a buyer  $i$  has positive demand in any period is equal to  $\pi_i = \alpha_{0i} / (\alpha_{0i} + \alpha_{1i})$ .

At the start of each period, some buyers and intermediaries are unmatched and directly offer bilateral contracts to unmatched producers. Let  $\mathcal{I}_{BS}$  denote the set of buyers who enter bilateral contracts that end when their demand switches to zero. For a buyer  $i \in \mathcal{I}_{BS}$ , the steady-state probability that they are unmatched is

$$v_i = (1 - \pi_i) \alpha_{0i}. \quad (10)$$

Let  $\mathcal{I}_{BI}$  be the set of buyers who directly contract with producers in contracts that continue when demand switches. Let  $\mathcal{I}_M$  be the set of buyers who contract with intermediaries. For both types of contracts, producers never separate. Therefore, for any buyer  $i \in \mathcal{I}_{BI} \cup \mathcal{I}_M$ , the steady-state probability that they are unmatched is  $v_i = 0$ . The total measure of new contracts offered in the producer market is given by

$$v = \int_{\mathcal{I}_{BS}} v_i dF. \quad (11)$$

The steady-state measure of producers in bilateral contracts,  $n_B$ , is given by

$$n_B = \int_{\mathcal{I}_{BS}} \pi_i dF + \int_{\mathcal{I}_{BI}} dF. \quad (12)$$

The steady-state measure of producers in intermediated contracts,  $n_M$ , is given by

$$n_M = \int_{\mathcal{I}_M} \pi_i dF. \quad (13)$$

The measure of unmatched producers after matching is

$$n_N = n - n_B - n_M. \quad (14)$$

The value of entering either bilateral or intermediated contracts is higher than the value of being unmatched, so there is an excess of producers who enter the producer market. This implies that  $n_N > 0$ .

Since matching is frictionless, all contracts offered in the producer market are immediately filled. The total measure of unmatched producers before matching,  $u$ , is given by

$$u = v + n_N. \quad (15)$$

Matching is random and contract offers are never made to matched producers, so the Bellman equation for an unmatched producer is given by

$$\bar{U} = \int_{\mathcal{I}_{BS}} \frac{v_i}{u} w_{BS}(\alpha_i) dF + \delta \bar{U}. \quad (16)$$

Note that the continuation value of being unmatched ( $\bar{U}$ ) consists of two components. The first term reflects the rent from matching with an unmatched buyer. The second term reflects the value of remaining unmatched.

To close the model, we solve for the steady-state number of producers that enter the economy. By assumption, producers enter the economy at the beginning of each period by paying an entry cost  $C$ . Entry drives down the likelihood that producers are matched, so they enter only until the continuation value of being unmatched in the labor market equals their entry cost. This yields the following condition:

$$\bar{U} = C. \quad (17)$$

We can now define an equilibrium in our economy.

**Definition 1.** *A steady-state equilibrium is a distribution of contracts offered by each buyer and intermediary such that:*

1. *All contracts are offerer-optimal, contract-specific, and stationary;*
2. *Each player's pre-matching continuation value is determined by steady-state transition probabilities and frictionless and random matching via Equation (16);*

3. The measure of producers in the economy is derived from the producer entry condition, given by Equation (17).

**Proposition 2.** *There exists a unique steady-state equilibrium. If  $y$  and  $C$  are sufficiently high and  $F$  has full support on  $[0, 1] \times [0, 1]$ , then a non-zero measure of buyers enter bilateral contracts, while another non-zero measure of buyers enter intermediated contracts.*

*Proof.* By Equation (17),  $\bar{U}$  equals the entry cost  $C$ . Given  $\bar{U}$ , we can compare the values of  $w_{BS}(\alpha_i)$ ,  $w_{BI}(\alpha_i)$ ,  $p(\alpha_i)$ , and  $y$  for each  $i$  using Equations (4), (5), and (9) to determine  $\mathcal{I}_{BS}$ ,  $\mathcal{I}_{BI}$ , and  $\mathcal{I}_M$ . Having derived these, we can obtain unique values for  $v_i$ ,  $v$ ,  $n_B$ , and  $n_M$  from Equations (10), (11), (12), and (13). We plug these into Equation (16) to solve for a unique value for  $u$ . Plugging  $u$  into Equations (14) and (15) then yields a unique value for  $n$ . The desired statement then follows from Proposition 1.  $\square$

## 4.2 Empirical Implications

Having shown that there exists a unique steady-state equilibrium in this exchange economy, we can now explore the model's predictions regarding equilibrium behavior.

### Effects of Intermediation on Producers

We first compare the pay, separation rates, and idleness of bilaterally contracted and intermediated producers in the steady-state equilibrium. We focus on the interesting case where all contract types coexist,<sup>20</sup> and derive four findings.

First, bilaterally contracted producers in our model have *higher average pay* per unit effort than intermediated producers. This follows from the fact that  $w_B(\alpha_i) > w_M$  for all buyers  $i$  with fluctuating demand.

Second, bilaterally contracted producers have *more dispersed pay* per unit effort than those of intermediated producers. This is because  $w_B(\alpha_i)$  takes on different values depending on  $\alpha_i$ , which is heterogeneous across buyers, while  $w_M$  is constant.

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<sup>20</sup>Specifically, we assume that  $y$ ,  $C$  and  $F$  are such that  $|\mathcal{I}_M|, |\mathcal{I}_{BI}|, |\mathcal{I}_{BS}| > 0$ . If  $F$  has full support on  $[0, 1] \times [0, 1]$ , then by Proposition 1 there exists  $y$  and  $C$  such that this holds.

Third, bilaterally contracted producers have *higher separation rates*. We define a producer's separation rate as the probability that a matched producer becomes unmatched at the start of the next period. This probability is higher for bilaterally contracted producers, since they may separate from buyers when their demand changes, while intermediated producers are always reallocated among the intermediary's clients.

Fourth, bilaterally contracted producers are *more likely to be idle* during their employment spells. We define idleness as the steady-state probability that a producer is matched but does not exert effort. In our model, intermediated producers are never idle, while bilaterally contracted producers may have positive idleness.

**Corollary 1.** *Compared to intermediated producers, bilaterally contracted producers have higher average pay, more dispersed pay, higher separation rates, and higher idleness in equilibrium.*

### Trade and Welfare with and without Intermediaries

We next analyze how the presence of intermediaries alters the welfare of buyers and producers. We find that intermediaries benefit some buyers, but on average reduce producer rents.

To show this, we consider two economies: one with intermediaries and one without. We assume that producers freely enter at some exogenous cost  $C$  in both economies. Therefore, the producer's continuation values when unmatched, and hence the contractual terms offered to them by buyers under bilateral contracting, are the same in the two economies. The only difference is the added possibility of intermediated contracts.

The introduction of intermediaries may cause three types of buyers to switch to intermediation: buyers who do not initially consume services, buyers who initially choose bilateral contracts with idleness, and buyers who initially choose bilateral contracts with separation. We denote the three subsets of switching buyers as  $\mathcal{S}_0$ ,  $\mathcal{S}_I$ , and  $\mathcal{S}_S$ .<sup>21</sup> The contracting choices of the remaining buyers are unchanged. We focus on the case where the measure of buyers switch from no consumption to intermediation and the measure of buyers switch from bilateral contracts to intermediation are

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<sup>21</sup>Assuming that indifferent buyers do not switch, we have that  $\mathcal{S}_0 = \{i \mid p(\alpha_i) < y < w_B(\alpha_i)\}$ ,  $\mathcal{S}_I = \{i \mid p(\alpha_i) < w_{BI}(\alpha_i) < \min\{y, w_{BS}(\alpha_i)\}\}$ , and  $\mathcal{S}_S = \{i \mid p(\alpha_i) < w_{BS}(\alpha_i) < \min\{y, w_{BI}(\alpha_i)\}\}$ .

both nonzero.<sup>22</sup>

It follows easily that the introduction of intermediaries benefit buyers with fluctuating demand, but it hurts incumbent producers who were initially earning pay premiums. By Corollary 1, the average producer pay per unit effort falls. Producer separation rates and idleness also fall. At the same time, the per-period payoff of buyers who switch to intermediation increases, while the per-period payoff of the remaining buyers are unchanged.

The introduction of intermediaries also unambiguously increases the measure of buyers who receive services. However, it is ambiguous whether the measure of matched producers ( $n_M + n_B$ ) increases or falls. On one hand, intermediaries enable more buyers to afford services, which increases demand for producers. On the other, intermediaries reduce idleness, so fewer producers are needed to fulfill the same level of demand. The relative magnitude of these two effects depends on the distribution of buyer types.

**Corollary 2.** *When intermediaries are introduced into an economy, the payoffs of some buyers increase, while the payoffs of the remaining buyers are unchanged. The mean pay per unit effort received by producers falls. The measure of buyers who receive services increases. The total measure of matched producers increases if and only if  $\int_{S_0} \pi_i dF > \int_{S_I} (1 - \pi_i) dF$ .*

### Intermediation is Limited by the Extent of the Market

An important feature of our model is that intermediation is efficiency-enhancing only if there are multiple sellers and buyers. Suppose, as an extreme case, that there is only one buyer and one producer in the economy. In such a case, it is never profitable for a buyer to use an intermediary, since the intermediary will require a markup and the cost savings from demand aggregation only emerge when the market is sufficiently big. Therefore, intermediation is more likely when *the extent of the market* — i.e. the available number of buyers and sellers — is large. This result is closely related to Stigler (1951), who conjectures that firms spin-off production stages because of increasing economies of scale as the market grows.

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<sup>22</sup>Formally, we assume that  $y$ ,  $C$  and  $F$  are such that  $|S_0| > 0$  and  $|S_I \cup S_S| > 0$ . This condition requires that there exists  $\alpha_i, \alpha'_i \in \text{supp } F$  such that  $y > w_B(\alpha_i) > p(\alpha_i)$  and  $w_B(\alpha'_i) > y > p(\alpha'_i)$ . If  $F$  has full support on  $[0, 1] \times [0, 1]$ , then by Proposition 1 there exists  $y$  and  $C$  such that this holds.



**Remark 1.** *In an economy with a single buyer and a single producer, intermediaries are always matched with zero buyers and zero producers.*

### 4.3 Exogenous producer population

So far we have assumed that the number of producers is determined via endogenous entry. In this section, we analyze an economy where the number of producers is fixed exogenously at  $n > 1$ .

We first show that if  $n$  is sufficiently large, then the producer market becomes very slack, buyers prefer to enter contracts with idleness, and there is no intermediation.

**Proposition 3.** *Assume the number of producers  $n$  is exogenously fixed. There exists a unique steady-state equilibrium. If  $n$  is sufficiently large, then there are no intermediated contracts.*

*Proof.* Given  $\bar{U}$ , we can compare the values of  $w_{BS}(\alpha_i)$ ,  $w_{BI}(\alpha_i)$ ,  $p(\alpha_i)$ , and  $y$  for each  $i$  using (4), (5), and (9) to determine  $\mathcal{I}_{BS}$ ,  $\mathcal{I}_{BI}$ , and  $\mathcal{I}_M$ . Having derived these, we can obtain unique values for  $v_i$ ,  $v$ ,  $n_B$ , and  $n_M$  from (10), (11), (12), and (13). Plugging  $w_{BS}(\alpha_i)$ ,  $w_{BI}(\alpha_i)$ , and  $w_M(\alpha_i)$  into (16) yields a contraction mapping. Therefore a unique fixed point for  $\bar{U}$  exists. By (14), (15), and (16),  $\bar{U}$  is decreasing in  $n$ , and  $\bar{U} \rightarrow 0$  as  $n \rightarrow \infty$ . The desired result follows by Proposition 1.  $\square$

We next compare economies with and without intermediaries assuming that  $n$  is fixed and that some nonzero measure of buyers choose intermediation when possible. In this economy, introducing intermediaries has both direct and indirect effects. The direct effect holds  $\bar{U}$  fixed and was characterized in the previous subsection. Due to this effect, various types of buyers switch to intermediation, producer rents fall, and the number of matched producers may or may not increase. Without free entry, however,  $\bar{U}$  also adjusts. A change in  $\bar{U}$  affects all of the endogenous variables in the model, so intermediation has indirect spillover effects onto all producers.

The sign of the resulting change in  $\bar{U}$  depends on the distribution of buyer types  $\alpha_i$ . If a large set of buyers switch from being unmatched to matched, then  $\bar{U}$  would increase due to reduced wait time in a tightened producer market. This indirect effect raises the pay of all producers. On the other hand, if no buyers switch from being unmatched to being matched, then  $\bar{U}$  would

unambiguously fall due to producer rent reductions associated with initially matched buyers switching to intermediation. This indirect effect instead reduces the pay of all producers.

## 5 Extensions

This section extends the model to study the relationship between intermediation, producer specialization, and intermediary reputation concerns. We first incorporate an endogenous choice by producers to specialize in different capabilities upon entry into the economy. Then, we add the possibility that poor performance by an intermediary is made known to a broader set of buyers.

### 5.1 Specialization

We consider a two-task economy with a measure of buyers  $i$  who have unit demand in every period, for either one of two tasks  $d_{it} \in \{A, B\}$ . Each buyer's demand switches from one task to another task with some symmetric probability  $\alpha_i$  at the start of each period.<sup>23</sup> There is also an excess measure of producers who choose whether to become either specialists in one of two activities needed by buyers or a generalist with middling skill in both services. Let  $\phi_j \in \{A, B, G\}$  denote the chosen type of the producer, where  $A$  and  $B$  refer to specialists and  $G$  refers to the generalist. The output depends on the buyer's demand  $d_{it}$ , the producer's type  $\phi_j$ , and the producer's chosen effort  $e_t$ , and is given by

$$y_{it} = \left[ y \cdot \mathbf{1}\{\phi_j = G\} + (y + \Delta_i) \cdot \mathbf{1}\{\phi_j = d_{it}\} \right] e_t,$$

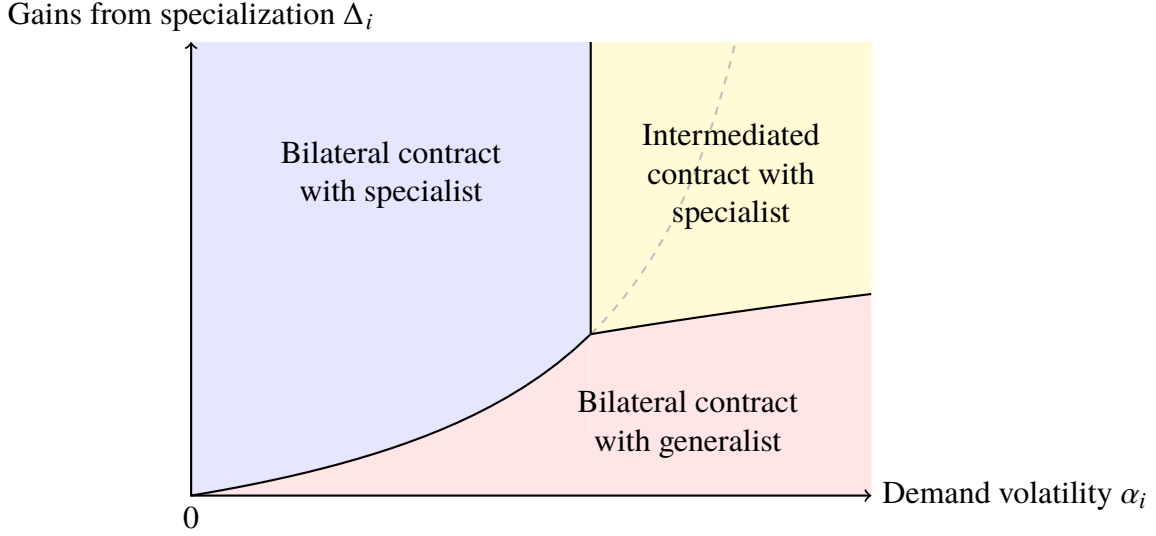
where  $\Delta_i > 0$  denotes the buyer-specific *gains from specialization*. If the specialist exerts effort, output is high when demand and the producer's type are well-matched, but low when they are not. Output is always middling for generalists who exert effort.

As before, neither pay  $w_t$  nor effort  $e_t$  are contractible and must be incentivized through relational contracts. We assume that output  $y$  is sufficiently large so that buyers are always able to receive positive profit by bilaterally contracting with a generalist, so there are never buyers

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<sup>23</sup>This is essentially simplifying the model in Section 2 by assuming  $\alpha_{Ai} = \alpha_{Bi} = \alpha_i$ .

Figure 4: Optimal contractual choice in the presence of gains from specialization



who do not receive services. We also assume that each producer's entry cost  $C$  is sufficiently large so that specialist producers never remain in a contract but become idle when the demand of their buyer changes. These assumptions allow us to focus on each buyer's choice between directly contracting with a generalist, directly contracting a specialist who is never idle, and intermediation. There are  $K$  intermediaries for each task. Each intermediary can only contract with producers specializing in that task and are randomly matched with entrepreneurs who offer intermediated contracts.<sup>24</sup>

For buyers with large  $\alpha_i$  and large  $\Delta_i$ , intermediation dominates either bilateral contracting arrangement. Due to double marginalization, however, it is optimal to directly contract with producers if demand is volatile and gains from specialization are small. It is also never optimal to enter an intermediated contract wherein the intermediary contracts with a generalist producer and sends the producer to different clients. Therefore, some buyer choose intermediated contracts with specialists, while others bilaterally contract with either specialists or generalists.

**Proposition 4.** *A unique steady-state equilibrium exists in a two-task economy. Buyer  $i$  chooses intermediated contracts if and only if their demand volatility  $\alpha_i$  and gains from specialization  $\Delta_i$  are both sufficiently large.*

<sup>24</sup>This setup implicitly assumes that each intermediary specializes in monitoring one type of task.

Another implication is that the presence of intermediaries encourages producers to specialize. In the absence of intermediaries, buyers with greater  $\alpha_i$  and hence more volatile demand contract with generalists, while those with smaller  $\alpha_i$  and hence less volatile demand contract with specialists. The dashed curve in Figure 4 shows the boundary between contracting with a specialist and with a generalist in the absence of intermediation. When intermediation becomes possible, however, their demand for directly contracted generalists is replaced with demand for intermediated specialists. This causes the overall demand for specialists to rise. In response, more producers choose to become specialists. Correspondingly, there are fewer bilateral contracts with elevated pay, so fewer producers enter, and the measure of unmatched producers falls. To show this formally, we assume that  $(\alpha_i, \Delta_i)$  are drawn from a distribution  $G$  that a non-zero measure of buyers choose intermediated contracts, and that the conditional distribution of  $\alpha_i$  given  $\Delta_i$  has positive support on  $[0, 1]$ .

**Corollary 3.** *When intermediaries are introduced into a two-task economy, the measure of specialist producers increase and the measure of unmatched producers falls.*

## 5.2 Reputation Concerns

Thus far, our model follows [Shapiro and Stiglitz \(1984\)](#) in assuming that producers are motivated to perform through the threat of contract termination, which would require them to wait to rematch in an anonymous matching market. Another source of motivation, as modeled by [Klein and Leffler \(1981\)](#), is that producers may lose reputational capital when they renege on promises to deliver high-quality services, causing a broader set of buyers to withhold future business.

In this subsection, we show that reputation concerns can play an important role in shaping the choice between intermediation and bilateral contracting. In particular, we show that intermediation becomes cheaper if the performance of the intermediary is partially observable to outside parties.

To see this formally, we consider a two-task economy with reputable intermediaries, where low effort by a producer is communicated with some probability to other buyers who can withhold future business from the producer's matched intermediary. Specifically, we assume that with probability  $\gamma \in [0, 1]$ , the effort choice by a producer contracted by the intermediary is observed

by another buyer, who is drawn among all buyers with uniform probability. The parameter  $\gamma$  measures the *ease of word-of-mouth communication* as enabled by communication technologies such as the Internet and social media platforms.<sup>25</sup>

As  $\gamma$  increases, the intermediary faces a harsher punishment if it reneges on its contracts. To see this, note that an intermediary's mean continuation value from being matched with an buyer is  $\tilde{V} = \frac{1}{|I|} \int_{I_0} \pi_i V_{1i} + (1 - \pi_i) V_{0i} dF > 0$ . When  $\gamma > 0$ , the intermediary's binding IC constraint is now given by:

$$V_1 \geq p_1 + \delta(-\gamma\tilde{V}), \quad (\text{M-IC-w'})$$

In other words, the threat of multilateral punishment means that a reduced mark-up is needed to incentivize the intermediary to perform. The unit cost of intermediated service therefore falls as  $\gamma$  increases.

As the cost of intermediation decreases, a larger measure of buyers choose intermediation. Correspondingly, the measure of buyers who directly contract with specialists and generalists both fall. This enables greater producer specialization, since total demand for specialized producers increases. At the same time, there are fewer bilateral contracts with elevated pay, so the measure of unmatched producers falls.

**Proposition 5.** *A unique steady-state equilibrium exists in a two-task economy with reputable intermediaries. As the ease of word-of-mouth communication increases, the measure of intermediated producers increases, the measure of specialist producers increases, and the measure of unmatched producers falls.*

**Remark.** By construction, our model disallows two possibilities that may in reality affect intermediation choices. First, producers themselves may build and maintain reputational capital. If so, they would not rematch on an *anonymous* market, as modeled above. Instead, if a producer shirks their contractual responsibilities, buyers can cause producers to face difficulty rematching thereafter. In this case, the producer faces a stronger incentive to perform and does so even if the

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<sup>25</sup>Here the intermediary will not wish to renege on more than one of its clients if she does not wish to do so for a single client, since an buyer may learn of bad service provided to multiple clients from word-of-mouth communication but can only punish maximally once.

particular buyer they currently provide services for is expected to have no demand next period. Therefore, intermediation becomes less attractive if producers maintain reputational capital.

Second, producers matched with the intermediary may be able to communicate with one another about the intermediary’s actions, such as through regular business conferences. This opens up the possibility that producers collectively punish intermediaries in response to contract infringement (a la [Levin 2002](#)). In our model, the threat of losing multiple producer relationships has no bite because intermediaries can immediately rematch with new producers. However, if rematching with producers is assumed to be difficult for intermediaries, then multilateral relational contracts between an intermediary and its producers will increase the intermediary’s incentive to perform, thereby lowering its service fee. Intermediation therefore becomes more attractive if producers can collectively punish intermediaries.

## 6 Applications

In this section, we show that the model’s prediction provide a unified explanation of the empirically observed drivers and effects of professional service outsourcing. We then use the model to think about retail franchises and online platforms.

### 6.1 Professional Service Outsourcing

To fix ideas, consider an entrepreneur who needs a service performed. She can fulfill the demand either by employing an in-house worker or contracting it out to an intermediary firm that allocates its employees to clients. Entrepreneurs may make these employ-or-outsource decisions for various professional services, including cleaning, security, accounting, legal, IT, and HR services.<sup>26</sup> These decisions determine the labor boundary of firms.

Our model provides a novel lens with which to understand these decisions as well as the function of professional service firms. We can define *employment* as a bilateral relational

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<sup>26</sup>Professional service firms account for 12 percent of U.S. employment and are among the largest employers in the world ([Berlingieri 2013](#)). More broadly defined, services account for the vast majority of cross-establishment trade ([Bostanci and Kambhampati 2022](#)).

contract between an employer and a worker and *outsourcing* as an intermediated contract in which an intermediary employs workers and assigns them to different entrepreneurs. Under these definitions, the model generates predictions for the determinants and effects of professional service outsourcing that are consistent with evidence.

For example, [Abraham and Taylor \(1996\)](#) find that establishments with cyclical demand and specialized needs are more likely to outsource accounting services. [Houseman \(2001\)](#) similarly finds that the need to accommodate fluctuations in workload is a commonly cited reason for using temp agencies. Consistent with this, Propositions 1 and 4 suggest that outsourcing is more likely when demand is volatile and gains from specialization are large.

Another frequent finding in the literature is that outsourced workers earn lower and more compressed wages ([Dube and Kaplan 2010](#); [Goldschmidt and Schmieder 2017](#); [Drenik et al. 2020](#)). Outsourced workers are also documented to have lower hazard into unemployment than comparable direct employees ([Guo, Li and Wong 2024](#)). These findings are consistent with Corollary 1, which shows that outsourced workers earn lower and less dispersed wages, and are less likely to become unmatched.

[Garicano and Hubbard \(2009\)](#) find that both the share of lawyers that specialize and the share of lawyers working in specialized firms increases with market size.<sup>27</sup> This finding confirms Remark 1 and Corollary 3, which show that market size increases intermediation, which in turn enables specialization.

Finally, [Bergeaud et al. \(2021\)](#) provide causal evidence that the rise of broadband internet increased outsourcing and increased the homogeneity of occupations within firms. Furthermore, macroeconomic studies indicate that firms are increasingly outsourcing their non-core activities, workers and firms have become increasingly specialized, and unemployment has fallen over the past half century ([Katz and Krueger 1999](#); [Weil 2014](#); [Handwerker 2023](#)). These trends may be partly explained by the arrival of communication technologies like the cellphone, the Internet, and social media, which have increased the role of branding and reputation in the economy. These trends are consistent with Proposition 5, which shows that ease of word-of-mouth communication increases intermediation and specialization and reduces unemployment.

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<sup>27</sup>For related evidence, see also [Baumgardner \(1988a,b\)](#) and [Duranton and Jayet \(2011\)](#).

## 6.2 Other Intermediary Organizations

There are many other examples of real-world intermediaries that our model can be used to understand. Table 1 provides a list. Here we discuss two particular examples.

**Franchises.** Franchisors are a large and growing class of intermediaries. All around the world, franchisors direct customers to local franchisees, who perform the actual services, and ensure that they fulfill a standard of performance. Because of this, consumers in many cities can buy coffee from either a local barista or a franchise such as Starbucks or Dunkin Donuts. They can get fast food from either local restaurants or franchises such as MacDonald’s or Subway. For a night’s rest, travelers can purchase hospitality either directly from a locally owned bed-and-breakfast or through a hotel franchise such as Marriott. For haircuts, fashionistas can visit either a local hairdresser or a franchise such as Supercuts.<sup>28</sup>

The franchisor can be viewed as an intermediary that maintains relational contracts with a large set of both franchisees and customers. They direct customers to franchised stores as customers travel to different locations and ensure that franchisees provide quality services to customers through the threat of contract termination.<sup>29</sup> Therefore, by Proposition 1, airports filled with itinerant travelers should feature more franchised stores, while small towns with immobile populations are more likely to have independent operations. Independent operations will also tend to be generalist and more susceptible to idleness, by Corollary 1 and Proposition 4.

**Online platforms.** With the rise of the Internet and smartphones, commuters can now choose between directly contracting with drivers or relying on ride hailing platforms like Uber. Shoppers can purchase goods through direct relationships with merchants or through e-commerce platforms such as Amazon and Alibaba. Owners can let properties either through direct relationships or through rental platforms such as Airbnb.

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<sup>28</sup>For a comprehensive empirical treatment of franchises and franchise contracts, see [Blair and Lafontaine \(2005\)](#).

<sup>29</sup>This view is supported by the website of the [International Franchise Association \(2024\)](#), a leading trade group, which explains that: “At its core, franchising is about the franchisor’s brand value, how the franchisor supports its franchisees, how the franchisee meets its obligations to deliver the products and services to the system’s brand standards and most importantly – franchising is about the relationship that the franchisor has with its franchisees.” See also [Klein \(1995\)](#), who provides a related repeated-game model of franchise contracts.



Table 1: Examples of relational intermediaries that monitor performance and direct allocations

| Intermediaries                                                                                   | Buyers     | Producers   |
|--------------------------------------------------------------------------------------------------|------------|-------------|
| Professional service firms<br>(e.g., law, accounting, HR, cleaning, security, consulting)        | Clients    | Workers     |
| Retailers, wholesalers, and e-commerce platforms<br>(e.g., Walmart, Amazon, Alibaba, eBay, Etsy) | Customers  | Sellers     |
| Franchisors<br>(e.g., Marriott, Starbucks, McDonald's, UPS)                                      | Consumers  | Franchisees |
| Global sourcing firms (e.g., Li & Fung)                                                          | Retailers  | Suppliers   |
| Ride-hailing platforms (e.g., Uber, Lyft, Grab, Didi)                                            | Riders     | Drivers     |
| Online rental markets (e.g., Airbnb, Turo)                                                       | Renters    | Owners      |
| Online labor markets (e.g., Upwork)                                                              | Businesses | Freelancers |
| Hospitals and clinics                                                                            | Patients   | Doctors     |
| Schools and universities                                                                         | Students   | Teachers    |

These online platforms can also be viewed as centralized intermediaries that aggregate demand, direct exchange, and monitor performance. For example, Uber operates feedback and dispatch systems to learn about performance from riders, direct drivers to serve riders as demand fluctuates, and incentivize performance through the promise of continued business. Similarly, e-commerce platforms operate recommendation and review systems that direct exchange and provide incentives for platform participants to perform. The function of these algorithmic computer systems is similar to the allocative and monitoring tasks traditionally performed by human managers in dispatch companies or professional service firms.<sup>30</sup> To incentivize these platforms to perform these functions, a markup is therefore required. By Proposition 1, an occasional rider should call for an Uber ride, but a company with persistent demand will prefer to directly employ a full-time driver instead to avoid the markup.

<sup>30</sup>Our view of online platforms as an incentive system has not received much attention in the existing economics literature. It instead focuses on platform pricing in the presence of usage and membership externalities (Rochet and Tirole 2003, 2006) and the impact of platforms on consumer search (Brynjolfsson and Smith 2000; Ellison and Ellison 2009). A notable exception is Hagi and Wright (2015), who model the organizational difference between vertical integration and multi-sided platforms as a difference in the allocation of decision rights, following Gibbons (2005).

## 7 Conclusion

The modern economy is riddled with intermediaries that maintain relational contracts on both sides of a market and coordinate transactions. In this paper, we theorize that these centralized monitors and coordinators exist to redress a problem of trust that pervades economic exchange and causes misallocation. We develop simple repeated-game models of intermediation in frictionless matching markets to formalize this idea. The models explain why and when the *visible hand* of monitoring intermediaries may replace the *invisible hand* of competitive markets in coordinating activities and allocating resources.

Our two main propositions show that buyers sort into decentralized exchange and monitor coordination in a unique steady-state equilibrium. Buyers with sufficiently volatile demand choose intermediation, while buyers with long-lasting demand choose bilateral contracts. The reason for this result is that intermediaries are useful only when demand is short-lived and a markup is needed to incentivize the intermediary to perform.

In extended models, we show that the optimal choice between intermediation and bilateral contracting depends on gains from specialization, market tightness, market size, and reputational effects. In contrast to models of intermediaries based on search frictions and asymmetric information, we show that falling communication costs can increase intermediation. Moreover, the effects of intermediation on output, welfare, the level of specialization, and the distribution of rents in the economy are characterized.

Our approach explains empirical findings regarding the determinants and effects of professional service outsourcing. It can also be used to understand the structure of supply chains, franchises, online platforms, and many other intermediary organizations. Given the tractability and applicability of our framework, further exploration informed by our approach may yield more insights concerning the role of intermediaries in the economy.

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## A Appendix

We omit the subscript  $i$  in the proofs. We first establish the following two lemmas.

**Lemma A.1.** *Suppose an optimal bilateral relational contract exists. Under this contract, if  $\beta > 0$ , then  $w_0 = 0$ , both (B-IC1) and (B-IC0) bind, and (P-IC1) is slack.*

*Proof.* Prove by contradiction. Suppose that  $w_0 > 0$ . Since there is an excess of producers in the frictionless matching market,  $\bar{\Pi}_1 = \Pi_1$ . So

$$\begin{aligned}\Pi_0 &= -w_0 + \delta \left[ \alpha_0 \Pi_1 + (1 - \alpha_0)(\beta \Pi_0 + (1 - \beta) \bar{\Pi}_0) \right] \\ &= -w_0 + \delta \left[ \alpha_0 \bar{\Pi}_1 + (1 - \alpha_0)(\beta \Pi_0 + (1 - \beta) \bar{\Pi}_0) \right].\end{aligned}$$

Also note that  $\bar{\Pi}_0 = \delta \left[ \alpha_0 \bar{\Pi}_1 + (1 - \alpha_0) \bar{\Pi}_0 \right]$ . Therefore,

$$\Pi_0 - \bar{\Pi}_0 = -w_0 + \delta(1 - \alpha_0)\beta(\Pi_0 - \bar{\Pi}_0) < \delta(1 - \alpha_0)\beta(\Pi_0 - \bar{\Pi}_0),$$

where the inequality comes from  $w_0 > 0$ . Since  $\delta \in (0, 1)$  and  $\beta \in (0, 1]$ , we discuss whether  $\alpha_0 = 1$ . If  $\alpha_0 < 1$ , the equation above cannot be satisfied, which contradicts that  $w_0 > 0$ . On the other hand, if  $\alpha_0 = 1$ ,  $\bar{\Pi}_0 = \delta \bar{\Pi}_1$  and then  $\Pi_0 = -w_0 + \delta \bar{\Pi}_1 < \bar{\Pi}_0$ , which contradicts to (B-IC0) and also indicates that  $w_0 = 0$ . Therefore,  $w_0 = 0$  and  $\Pi_0 = \bar{\Pi}_0$ . Thus both (B-IC1) and (B-IC0) bind.

We now show that (P-IC1) is slack. Suppose it binds, namely  $U_1 = \bar{U}$ . Then given  $w_0 = 0$  and  $\delta < 1$ , plug in  $U_1 = \bar{U}$  and get

$$\begin{aligned}U_0 &= \delta \left[ \alpha_0 \bar{U} + (1 - \alpha_0)(\beta U_0 + (1 - \beta) \bar{U}) \right] \\ &< (1 - \alpha_0)\beta U_0 + (1 - (1 - \alpha_0)\beta) \bar{U}.\end{aligned}$$

The inequality above indicates that  $U_0 < \bar{U}$ , which contradicts to (P-IC0). So (P-IC1) is slack.  $\square$

**Lemma A.2.** *For any buyer who directly contracts with producers, maximizing  $\Pi_1$  is equivalent to minimizing  $w_1$ .*

*Proof.* Based on Lemma A.1, the buyer's continuation payoffs can be written as

$$\Pi_1 = \frac{1 - \delta(1 - \alpha_0)}{1 - \delta\alpha_0\alpha_1 - \delta(2 - \alpha_0 - \alpha_1) + \delta^2(1 - \alpha_0)(1 - \alpha_1)}(y - w_1),$$

$$\Pi_0 = \frac{\delta\alpha_0}{1 - \delta(1 - \alpha_0)}\Pi_1.$$

Since  $\frac{\partial \Pi_1}{\partial w_1} < 0$ , a buyer's problem is equivalent to minimize  $w_1$  subject to producer's incentive constraints.  $\square$

## Proof of Lemma 1

For simplicity, we write  $w_1$  as  $w$  in the rest of the proof. We complete the proof by analyzing and comparing the terms in the optimal bilateral relational contracts when choosing  $\beta = 0$  or  $\beta > 0$ .

**Choice 1:**  $\beta = 0$ . If the optimal choice of  $\beta$  is 0,  $U_1 = w - c + \delta[(1 - \alpha_1)U_1 + \alpha_1\bar{U}]$ . The buyer optimally chooses  $w$  subject to a binding (P-IC-e), which gives

$$w_{BS} = \frac{1}{\delta(1 - \alpha_1)}c + (1 - \delta)\bar{U}.$$

**Choice 2:**  $\beta \in (0, 1]$ . If the optimal choice of  $\beta$  is greater than 0, a buyer's optimization problem becomes  $\min_{w, \beta} w$  subject to (P-IC-e), (P-IC0), (1), (2),  $0 < \beta \leq 1$ , and  $w \geq 0$ .

The Lagrangian is given by

$$\begin{aligned} L = & w + \lambda_1(U_1 - (w - c + \delta((1 - \alpha_1) \cdot U_1 + \alpha_1(\beta U_0 + (1 - \beta)\bar{U})))) \\ & + \lambda_2(U_0 - \delta(\alpha_0 \cdot U_1 + (1 - \alpha_0) \cdot (\beta U_0 + (1 - \beta)\bar{U}))) \\ & + \mu_1(w + \delta\bar{U} - U_1) + \mu_2(\bar{U} - U_0) + \mu_3(\beta - 1) + \mu_4(-w). \end{aligned}$$

The Kuhn-Tucker conditions are given by

$$\frac{\partial L}{\partial w} = 1 + \mu_1(1 - \frac{\partial U_1}{\partial w}) - \mu_2 \frac{\partial U_0}{\partial w} - \mu_4 \leq 0,$$

$$\begin{aligned}
\frac{\partial L}{\partial w} w &= 0, \\
\frac{\partial L}{\partial \beta} &= -\mu_1 \frac{\partial U_1}{\partial \beta} - \mu_2 \frac{\partial U_0}{\partial \beta} - \mu_3 = 0, \\
\lambda_1, \lambda_2 &> 0, \\
\mu_1, \mu_2, \mu_3, \mu_4 &\geq 0, \\
\mu_1(w + \delta \bar{U} - U_1) &= 0, \\
\mu_2(\bar{U} - U_0) &= 0, \\
\mu_3(1 - \beta) &= 0, \\
\mu_4 w &= 0.
\end{aligned}$$

Meanwhile, by taking derivatives on both sides of equations (1) and (2) with respect to  $w$ , get

$$\frac{\partial U_1}{\partial w} = 1 + \delta \left[ (1 - \alpha_1) \frac{\partial U_1}{\partial w} + \alpha_1 \beta \frac{\partial U_0}{\partial w} \right], \quad (\text{A1})$$

$$\frac{\partial U_0}{\partial w} = \delta \left[ \alpha_0 \frac{\partial U_1}{\partial w} + (1 - \alpha_0) \beta \frac{\partial U_0}{\partial w} \right]. \quad (\text{A2})$$

By taking derivatives on both sides of equations (1) and (2) with respect to  $\beta$ , get

$$\frac{\partial U_1}{\partial \beta} = \delta \left[ (1 - \alpha_1) \frac{\partial U_1}{\partial \beta} + \alpha_1 (U_0 - \bar{U} + \beta \frac{\partial U_0}{\partial \beta}) \right], \quad (\text{A3})$$

$$\frac{\partial U_0}{\partial \beta} = \delta \left[ \alpha_0 \frac{\partial U_1}{\partial \beta} + (1 - \alpha_0) (U_0 - \bar{U} + \beta \frac{\partial U_0}{\partial \beta}) \right]. \quad (\text{A4})$$

We proceed by the following steps.

**Step 1: Show that  $\mu_4 = 0$  and  $w > 0$ .** Suppose that  $w = 0$ . Then

$$\begin{aligned}
U_1 &= -c + \delta \left[ (1 - \alpha_1) U_1 + \alpha_1 (\beta U_0 + (1 - \beta) \bar{U}) \right] \\
&< (1 - \alpha_1) U_1 + \alpha_1 (\beta U_0 + (1 - \beta) \bar{U}),
\end{aligned}$$

where the inequality comes from that  $c > 0$  and  $\delta < 1$ . The inequality above implies that

$$U_1 < \beta U_0 + (1 - \beta)\bar{U} < U_0,$$

where the second inequality comes from (P-IC0). Meanwhile,

$$\begin{aligned} U_0 &= \delta \left[ \alpha_0 U_1 + (1 - \alpha_0)(\beta U_0 + (1 - \beta)\bar{U}) \right] \\ &< \alpha_0 U_1 + (1 - \alpha_0)(\beta U_0 + (1 - \beta)\bar{U}) \\ &< U_0, \end{aligned}$$

where the first inequality comes from  $\delta < 1$  and the second inequality comes from (P-IC0) and  $U_1 < U_0$ . Since  $U_0 < U_0$  can never be true, we know that  $w > 0$  and thus  $\mu_4 = 0$ .

The implication for  $w > 0$  is that

$$\frac{\partial L}{\partial w} = 1 + \mu_1 \left(1 - \frac{\partial U_1}{\partial w}\right) - \mu_2 \frac{\partial U_0}{\partial w} = 0,$$

**Step 2: Discuss the values of  $\mu_1$  and  $\mu_2$ .** **Case 1)**  $\mu_1 = \mu_2 = 0$ . In this case,  $\frac{\partial L}{\partial w} = 0$  is violated, indicating that this case is not possible. In other words, at least one of two incentive compatibility constraints bind.

**Case 2)**  $\mu_1 = 0$  and  $\mu_2 > 0$ . In this case,  $U_1 > w + \delta\bar{U}$  and  $U_0 = \bar{U}$ . Solve  $U_1$  and  $w$  based on equation (1) and get

$$\begin{aligned} U_1 &= \frac{1 - \delta(1 - \alpha_0)}{\delta\alpha_0} \bar{U}, \\ w &= c + \frac{(1 - \delta)^2 - \delta(1 - \delta)(\alpha_0 + \alpha_1)}{\delta\alpha_0} \bar{U}. \end{aligned}$$

So  $\frac{\partial U_1}{\partial \beta} = \frac{\partial U_0}{\partial \beta} = 0$ . Meanwhile,  $\frac{\partial L}{\partial w} = 0$  and  $\mu_1 = 0$  imply that  $1 = \mu_2 \frac{\partial U_0}{\partial w}$ .  $\frac{\partial L}{\partial \beta} = 0$  implies  $\mu_3 = 0$ , namely  $\beta < 1$ .

There are two conditions that need to be satisfied for this case to be feasible and optimal. First, for feasibility, the solved  $w$  and  $U_1$  need to satisfy  $U_1 > w + \delta\bar{U}$ . After plugging in  $U_1$  and

$w$  as functions of  $\bar{U}$ , this requires that

$$\frac{(1-\delta)\bar{U}}{c} > \frac{\alpha_0}{1-\alpha_1}.$$

Second, since we are considering the case where choosing  $\beta > 0$  is weakly better than choosing  $\beta = 0$ , the solved  $w$  needs to be lower than  $w_{BS}$ , which gives

$$\frac{(1-\delta)\bar{U}}{c} \leq \frac{\alpha_0}{1-\alpha_1}.$$

These two conditions contract each other, indicating that this case is not possible.

**Case 3)**  $\mu_1 > 0$  and  $\mu_2 > 0$ . In this case, both incentive compatibility constraints bind, which give  $U_1 = w + \delta\bar{U}$  and  $U_0 = \bar{U}$ . Under binding (P-IC-e) and (P-IC0), equations (1) and (2) can be satisfied only if

$$\frac{(1-\delta)\bar{U}}{c} = \frac{\alpha_0}{1-\alpha_1}.$$

In that case, the solved  $w$  also coincides with  $w_{BS}$ , and a buyer is thus indifferent among choosing any value of  $\beta \in [0, 1]$ .

**Case 4)**  $\mu_1 > 0$  and  $\mu_2 = 0$ . In this case,  $U_1 = w + \delta\bar{U}$ ,  $U_0 > \bar{U}$ ,  $\frac{\partial L}{\partial w} = 1 + \mu_1(1 - \frac{\partial U_1}{\partial w}) = 0$ , and  $\frac{\partial L}{\partial \beta} = -(\mu_1 \frac{\partial U_1}{\partial \beta} + \mu_3) = 0$ .

We discuss whether  $\beta = 1$  or not. If  $\beta \neq 1$ ,  $\mu_3 = 0$ , then  $\frac{\partial U_1}{\partial \beta} = 0$ . Given that, equation (A3) suggests that  $\frac{\partial U_0}{\partial \beta} < 0$ , while equation (A4) suggests that  $\frac{\partial U_0}{\partial \beta} > 0$ . Therefore, a contradiction exists, indicating that  $\beta = 1$ .

Given  $\beta = 1$  and  $\mu_3 > 0$ , solve

$$w_{BI} = \left( \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1) + \frac{\delta}{1-\delta}\alpha_0} \right) c + \left( \frac{1 + \frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1) + \frac{\delta}{1-\delta}\alpha_0} - \delta \right) \bar{U}.$$

In sum, if a buyer decides to choose  $\beta > 0$ , she will optimally choose  $\beta = 1$  with paying  $w_{BI}$  when her demand is 1. The comparison between choice 1 and choice 2 hinge on comparing  $w_{BS}$

and  $w_{BS}$ . It turns out that  $w_{BS} < w_{BI}$  if and only if

$$\frac{(1 - \delta)\bar{U}}{c} > \frac{\alpha_0}{1 - \alpha_1}.$$

Therefore, whenever the condition above holds, the buyer chooses  $\beta = 0$  (separating from the producer when demand becomes 0) and pays  $w_{BS}$  when her demand is 1. Otherwise, she chooses  $\beta = 1$  (retaining the producer when demand becomes 0) and pays  $w_{BI}$  when her demand is 1.

## Proof of Lemma 2

Observe that the continuation payoffs and incentive compatibility constraints for an intermediary in an intermediated contract are similar with those for a producer in a bilateral contract. The only differences are that the “cost of production” for an intermediary is  $w_M$  and its continuation value after separating from a buyer, given by  $\bar{V}$ , is 0. Since

$$\frac{(1 - \delta)\bar{V}}{w_M} \leq \frac{\alpha_0}{1 - \alpha_1}$$

for any  $\alpha_0$  and  $\alpha_1$ . Therefore, the value of  $p$  is determined by replacing  $c$  with  $w_M$  and  $\bar{U}$  with 0 in  $w_{BI}$ .

## Proof of Proposition 1

We prove the proposition in four steps.

**Step 1:** Compare bilateral contracting with separation with bilateral contracting with idleness.

The following lemma comes directly from Lemma 1.

**Lemma A.3.** *If  $\bar{U} > \bar{U}^{ei} \equiv \frac{c}{1-\delta}\alpha_0$ , there exists a unique cutoff  $\bar{\alpha}_1^{ei} \in (0, 1)$  such that  $w_{BS} < w_{BI}$  if and only if  $\alpha_1 < \bar{\alpha}_1^{ei}$ .*

**Step 2:** Compare bilateral contracting with separation with intermediated contracting.

**Lemma A.4.** *There exists a unique cutoff  $\bar{\alpha}_1^{eo} \in (0, 1]$  such that  $w_{BS} < p$  if and only if  $\alpha_1 < \bar{\alpha}_1^{eo}$ .*

*Proof.* Observe that  $w_{BS} < p$  if and only if

$$\frac{1}{\delta} \frac{1}{1 - \alpha_1} c + (1 - \delta) \bar{U} < \left( \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_0}{(1 - \alpha_1) + \frac{\delta}{1-\delta} \alpha_0} \right) \frac{c}{\delta} + \left( \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_0}{(1 - \alpha_1) + \frac{\delta}{1-\delta} \alpha_0} \right) (1 - \delta) \bar{U},$$

or

$$\frac{1}{1 - \alpha_1} \frac{c}{\delta} + (1 - \delta) \bar{U} - \frac{1}{\delta - \frac{\delta}{1 + \frac{\delta}{1-\delta} \alpha_0} \alpha_1} \left( \frac{c}{\delta} + (1 - \delta) \bar{U} \right) < 0.$$

Let the LHS to be  $f(\alpha_1) = \frac{1}{1 - \alpha_1} \frac{c}{\delta} + (1 - \delta) \bar{U} - \frac{1}{\delta - \frac{\delta}{1 + \frac{\delta}{1-\delta} \alpha_0} \alpha_1} \left( \frac{c}{\delta} + (1 - \delta) \bar{U} \right)$ . Observe that, when  $\alpha_0 = 0$ ,  $f(\alpha_1)|_{\alpha_0=0} = (1 - 1/\delta) \frac{c}{\delta} \frac{1}{1 - \alpha_1} + (1 - \frac{1}{\delta(1 - \alpha_1)}) (1 - \delta) \bar{U} < 0$ . In this case,  $w_{BS} < p$  for sure, so  $\bar{\alpha}_1^{eo} = 1$ .

When  $\alpha_0 > 0$ , observe that  $f(0) = (1 - \frac{1}{\delta}) \kappa < 0$ , and  $f(1)$  goes to infinity. We now show that the continuous function  $f(\alpha_1)$  intersects with 0 only once. Compute

$$\frac{\partial f(\alpha_1)}{\partial \alpha_1} = \frac{\zeta(\zeta \frac{c}{\delta} - \kappa) \alpha_1^2 - 2(c - \kappa) \zeta \alpha_1 + (\delta c - \zeta \kappa)}{(1 - \alpha_1)^2 (\delta - \zeta \alpha_1)^2}.$$

where  $\zeta = \frac{\delta}{1 + \frac{\delta}{1-\delta} \alpha_0}$  and  $\kappa = \frac{c}{\delta} + (1 - \delta) \bar{U}$ .

Observe that the numerator is a quadratic equation, where the coefficient of  $\alpha_1^2$  is negative, the coefficient of  $\alpha_1$  is positive, and the constant  $\delta c - \zeta(\frac{c}{\delta} + (1 - \delta) \bar{U})$  can be positive or negative. Therefore,  $f(\alpha_1)$  is either strictly increasing, or is first decreasing then increasing. In either case,  $f(\alpha_1)$  intersects with 0, with the intersect being  $\bar{\alpha}_1^{eo} \in (0, 1)$ .

In sum, there exists a unique cutoff  $\bar{\alpha}_1^{eo} \in (0, 1]$  such that  $w_{BS} < p$  if and only if  $\alpha_1 < \bar{\alpha}_1^{eo}$ .  $\square$

**Step 3:** Compare bilateral contracting with idleness with intermediated contracting.

**Lemma A.5.** If  $\bar{U} > \bar{U}^{io} \equiv \frac{(1 - \delta(1 - \alpha_0))c}{\delta(\delta(2 - (1 - \delta)\alpha_0) - 1)}$ , there exists a unique cutoff  $\bar{\alpha}_1^{io} \in (0, 1)$  such that  $w_{BI} < p$  if and only if  $\alpha_1 < \bar{\alpha}_1^{io}$ . Otherwise,  $w_{BI} < p$  for sure.

*Proof.* Observe that  $w_{BI} < p$  if and only if

$$\left( \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_0}{(1 - \alpha_1) + \frac{\delta}{1-\delta} \alpha_0} \right) c + \left( \frac{1 + \frac{\delta}{1-\delta} \alpha_0}{(1 - \alpha_1) + \frac{\delta}{1-\delta} \alpha_0} - \delta \right) \bar{U} < \left( \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_0}{(1 - \alpha_1) + \frac{\delta}{1-\delta} \alpha_0} \right) \left( \frac{c}{\delta} + (1 - \delta) \bar{U} \right),$$



or

$$\frac{1 + \frac{\delta}{1-\delta}\alpha_0}{1 + \frac{\delta}{1-\delta}\alpha_0 - \alpha_1} \left( \frac{2\delta-1}{\delta}\bar{U} - \frac{1-\delta}{\delta}\frac{c}{\delta} \right) < \delta\bar{U}.$$

Let the LHS to be  $g(\alpha_1) = \frac{1 + \frac{\delta}{1-\delta}\alpha_0}{1 + \frac{\delta}{1-\delta}\alpha_0 - \alpha_1} \left( \frac{2\delta-1}{\delta}\bar{U} - \frac{1-\delta}{\delta}\frac{c}{\delta} \right)$ . Observe that  $g(0) = \frac{2\delta-1}{\delta}\bar{U} - \frac{1-\delta}{\delta}\frac{c}{\delta}$  and  $g(1) = \frac{1 + \frac{\delta}{1-\delta}\alpha_0}{\frac{\delta}{1-\delta}\alpha_0} \left( \frac{2\delta-1}{\delta}\bar{U} - \frac{1-\delta}{\delta}\frac{c}{\delta} \right)$ . Also observe that  $g(0) - \delta\bar{U} = \frac{-(1-\delta)^2}{\delta}\bar{U} - \frac{1-\delta}{\delta}\frac{c}{\delta} < 0$ .

If  $\bar{U} < \bar{U}^{io}$ ,  $g(1) < \delta\bar{U}$  and thus the inequality holds for sure. Otherwise, since  $g(\alpha_1)$  is strictly increasing in  $\alpha_1$ , there exists a unique cutoff  $\bar{\alpha}_1^{io}$  by the intermediate value theorem.  $\square$

**Step 4:** Let  $\bar{U}^* = \bar{U}^{io}$  and  $\alpha^* = \min\{\bar{\alpha}_1^{io}, \bar{\alpha}_1^{eo}\}$ .

If  $\bar{U} < \bar{U}^{io}$ ,  $w_{BI} < p$  by Lemma A.5. Otherwise, if  $\bar{U} \geq \bar{U}^{io}$  there are two cases. If  $w_{BS} < w_{BI}$ , intermediated contract is optimal when  $p < w_{BS}$ , namely when  $\alpha_1 > \bar{\alpha}_1^{eo}$ . If  $w_{BS} > w_{BI}$  intermediated contract is optimal when  $p < w_{BI}$ , namely when  $\alpha_1 > \bar{\alpha}_1^{io}$ .

## Proof of Corollary 1

First, note that  $w_B(\alpha_i) > w_M$  for all  $i \in \mathcal{I}_{BS} \cup \mathcal{I}_{BI}$ . Therefore,  $E[w_B(\alpha_i) \mid i \in \mathcal{I}_{BS} \cup \mathcal{I}_{BI}] > E[w_M \mid i \in \mathcal{I}_M]$ . Second, note that  $w_B(\alpha_i)$  takes on different values depending on  $\alpha_i$ , which is heterogeneous across buyers, while  $w_M$  is constant. Therefore,  $\text{Var}[w_B(\alpha_i) \mid i \in \mathcal{I}_{BS} \cup \mathcal{I}_{BI}] > \text{Var}[w_M \mid i \in \mathcal{I}_M] = 0$ . Third, the separation rate is given by  $\beta_i \alpha_{1i}$ . Note that  $\beta_i = 1$  for all  $i \in \mathcal{I}_{BI}$ , while  $\beta_i = 0$  for all  $i \in \mathcal{I}_{BS} \cup \mathcal{I}_M$ . Fourth, the idleness for a buyer  $i \in \mathcal{I}_{BI}$  is given by  $(1 - \pi_i)(1 - \omega_i) > 0$ . By contrast, for  $i \in \mathcal{I}_{BS} \cup \mathcal{I}_M$ , idleness is zero.

## Proof of Corollary 2

With the introduction of intermediaries, the per-period payoff of buyer  $i \in \mathcal{S}_0$  increases by  $y - p(\alpha_i)$ . The per-period payoff of buyer  $i \in \mathcal{S}_I$  increases by  $w_{BI}(\alpha_i) - p(\alpha_i)$ . The per-period payoff of buyer  $i \in \mathcal{S}_S$  increases by  $w_{BS}(\alpha_i) - p(\alpha_i)$ . The per-period payoff of the remaining buyers are unchanged.

The average producer pay per unit effort falls, since  $w_B(\alpha_i) > w_M$  for all  $i \in \mathcal{S}_I \cup \mathcal{S}_S$ . Producer separation rates and idleness also fall, since buyers switch from bilateral contracts with

either idleness and separation to intermediated contracts in which buyers are never idle and never separate.

The measure of buyers who receive services increases by  $|\mathcal{S}_0|$ . The total measure of services provided in the economy increases by  $\int_{\mathcal{S}_0} \pi_i dF$ . However, for  $i \in \mathcal{S}_I$ , the steady-state measure of producers matched to these buyers fall, since idleness falls. The total measure of matched producer ( $n_M + n_B$ ) therefore increases if and only if  $\int_{\mathcal{S}_0} \pi_i dF > \int_{\mathcal{S}_I} (1 - \pi_i) dF$ .

## Proof of Proposition 4

Based on Lemma 1 and 2, the payments by buyer  $i$  under bilateral contracts or intermediation are

$$w_B(\alpha_i) = \frac{1}{\delta} \frac{1}{1 - \alpha_i} c + (1 - \delta) \bar{U},$$

$$w_G = w_M = \frac{c}{\delta} + (1 - \delta) \bar{U},$$

$$p(\alpha_i) = \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_i}{1 + \frac{\delta}{1-\delta} \alpha_i - \alpha_i} \left[ \frac{c}{\delta} + (1 - \delta) \bar{U} \right],$$

where  $w_B(\alpha_i)$  is the pay for a specialist,  $w_G$  is the pay for a generalist,  $w_M$  is the pay from an intermediary to an intermediated producer, and  $p(\alpha_i)$  is the service fee for intermediation.

The buyer's post-matching continuation payoffs when she has demand when choosing to directly contract with specialists, to directly contract with a generalist, or to outsource are thus

$$\Pi_B(\alpha_i) = \frac{y + \Delta_i}{1 - \delta} - \left[ \bar{U} + \frac{c}{(1 - \alpha_i)\delta(1 - \delta)} \right],$$

$$\Pi_G = \frac{y}{1 - \delta} - \left[ \bar{U} + \frac{c}{\delta(1 - \delta)} \right].$$

$$\Pi_M(\alpha_i) = \frac{y + \Delta_i}{1 - \delta} - \left[ \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_i}{1 + \frac{\delta}{1-\delta} \alpha_i - \alpha_i} \left( \bar{U} + \frac{c}{\delta(1 - \delta)} \right) \right],$$

respectively.

Three observations follow. First, a buyer prefers to bilaterally contract with a specialist than

bilaterally contract with a generalist if and only if  $\Pi_B(\alpha_i) \geq \Pi_G$ , or,

$$\Delta_i \geq \frac{\alpha_i}{1 - \alpha_i} \frac{c}{\delta}.$$

Second, a buyer prefers to bilaterally contract with specialists than intermediate if and only if

$$\left[ \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_i}{1 + \frac{\delta}{1-\delta} \alpha_i - \alpha_i} - 1 \right] \left[ \bar{U} + \frac{c}{\delta(1-\delta)} \right] \geq \frac{\alpha_i}{1 - \alpha_i} \frac{c}{\delta(1-\delta)}.$$

By Lemma A.3, there exists a unique cutoff  $\alpha^{ei}$  such that the buyer prefers to intermediate if and only if  $\alpha_i > \alpha^{ei}$ . Third, a buyer prefers an intermediated contract over bilateral contract with a generalist if and only if

$$\Delta_i \geq \bar{\Delta}(\alpha_i) \equiv \left[ \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_i}{1 + \frac{\delta}{1-\delta} \alpha_i - \alpha_i} - 1 \right] \left[ \bar{U} + \frac{c}{\delta(1-\delta)} \right].$$

Therefore, the buyer choose to intermediate if and only if  $\alpha_i$  and  $\Delta_i$  are both large enough.

### Proof of Corollary 3

Let  $\mathcal{S}_G$  denote buyers who switch from bilateral contracts with generalists to intermediated contracts with specialists when intermediaries become available. Let  $\mathcal{S}_B$  be the set of buyers who switch from bilateral contracts with specialists to intermediated contracts with specialists. Given the assumption on the distribution of  $(\Delta_i, \alpha_i)$ ,  $|\mathcal{S}_G|, |\mathcal{S}_B| > 0$ . The contractual choice of the remaining buyers are unchanged. Therefore, the measure of specialists unambiguously increase with intermediation.

Note that the Bellman equation for unmatched producers can be rewritten as

$$\bar{U} = \frac{v}{u} \frac{E[v_i U_{1i}]}{v} + \left( 1 - \frac{v}{u} \right) \delta \bar{U}.$$

With the introduction of intermediaries, more producers are contracted as specialists under intermediated contracts without separation, so  $v$  and  $E[v_i U_{1i}]$  both fall. This implies that  $\frac{v}{u}$

increases. It follows that the measure of unmatched producers  $n_N = u - v$  falls.

## Proof of Proposition 5

In a two-task economy with reputable intermediaries, the service fee is given by

$$p(\alpha_i) = \left( \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_i}{(1 - \alpha_i) + \frac{\delta}{1-\delta} \alpha_i} \right) w_M + \left( \frac{1 + \frac{\delta}{1-\delta} \alpha_i}{(1 - \alpha_i) + \frac{\delta}{1-\delta} \alpha_i} - \delta \right) (-\gamma \tilde{V})$$

This implies that

$$\Pi_M(\alpha_i) = \frac{y + \Delta_i}{1 - \delta} - \left[ \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta} \alpha_i}{1 + \frac{\delta}{1-\delta} \alpha_i - \alpha_i} (\bar{U} + \frac{c}{\delta(1 - \delta)}) \right] + \left( \frac{1 + \frac{\delta}{1-\delta} \alpha_i}{(1 - \alpha_i) + \frac{\delta}{1-\delta} \alpha_i} - \delta \right) \frac{\gamma \tilde{V}}{1 - \delta},$$

As  $\gamma$  increases,  $\Pi_M(\alpha_i)$  rises, so buyers switch to intermediated contracts with specialists from bilateral contracts with both generalists and specialists. Therefore, by the same logic as Corollary 3, the measure of specialists increases, while the measure of unmatched producers decreases.